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Dopamine-Inspired Reinforcement Learning for Improved Exploration

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Dopamine-Inspired Reinforcement Learning for Improved Exploration

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I hereby declare that this dissertation is all my own work, except as indicated in the text:

Signature _

A handwritten signature in black ink, appearing to be 'Sharifah', is placed over the signature line.

Date 28/04/2026

Abstract

Efficient exploration remains a fundamental challenge in reinforcement learning (RL), particularly in environments where standard temporal-difference (TD) learning and fixed exploration schedules often fail to discover informative states within practical training budgets. This project introduces a “bio-RPE” framework inspired by the dopaminergic functions of the biological brain, unifying temporal-difference (TD) error, Random Distillation (RND)-based novelty, and ensemble-driven uncertainty into a single adaptive signal. Built upon a Bootstrapped DQN, the framework employs three integration schemes (additive, multiplicative, and gated) to modulate learning rates and exploration dynamically. The framework was evaluated across three OpenAI Gymnasium environments spanning dense and sparse reward settings: CartPole-v1, MountainCar-v0, and Acrobot-v1, using six ablation conditions and 180 total training runs. Cross-environment findings were strongly environment-dependent: GATE emerged as the most consistent all-round performer, FULL achieved the strongest convergence stability and best overall performance on Acrobot, and novelty-enabled conditions directionally improved sparse exploration on MountainCar. Internal signal analysis revealed that bio-RPE trajectories functioned as an emergent convergence diagnostic, while RND novelty acted primarily as an early-training ignition signal due to rapid saturation. Although most pairwise differences were not statistically significant at $n = 10$ seeds, the framework showed meaningful directional improvements in exploration robustness and convergence stability while maintaining qualitative correspondence with several dopaminergic principles, including novelty habituation, gain modulation, and adaptive plasticity. This work demonstrates that biologically grounded signal integration successfully bridges neuroscience and computational RL, offering a robust path for improving agent stability in complex reward landscapes.

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1. Introduction

Reinforcement learning (RL) enables agents to discover optimal behaviours through trial-and-error interactions with an environment, formalised as Markov Decision Processes (MDPs) in which agents maximise expected cumulative reward (Sutton & Barto, 2018). Over the past decade, deep RL has achieved landmark results across game playing, robotic control, and sequential decision-making. Algorithms such as Deep Q-Networks (DQN) and policy-gradient methods have demonstrated the ability to solve complex tasks when reward signals are frequent and well-shaped (Mnih et al., 2015; Schulman et al., 2017).

Despite these advances, a persistent limitation of modern RL systems lies in inefficient exploration, particularly in environments with sparse rewards, delayed feedback, or deceptive local optima. In such settings, agents may fail to encounter informative rewards within reasonable training budgets, leading to slow convergence, high variance across seeds, or complete learning failure (Bellemare et al., 2016; Ecoffet et al., 2021). Benchmarks such as MountainCar-v0 and Acrobot-v1 illustrate this vividly: both provide near-constant negative rewards with no gradient to guide early exploration, exposing how standard strategies such as ϵ -greedy action selection or entropy regularisation provide insufficient guidance and motivate research into richer exploration mechanisms.

In parallel, neuroscience offers a compelling blueprint for more adaptive learning. Midbrain dopaminergic neurons do not merely signal scalar reward discrepancies; rather, they encode a rich, multifaceted reward prediction error (RPE) that combines novelty, uncertainty, surprise, and environmental volatility (Schultz, Dayan, & Montague, 1997; Schultz, 2016). This composite dopamine signal dynamically modulates synaptic plasticity, accelerating learning during high-surprise episodes, and also governs the exploration-exploitation trade-off via phasic and tonic components. Recent empirical and computational work reinforces this picture: Wang et al. (2024) demonstrate that transient dopamine responses to novelty facilitate efficient uncertainty-driven exploration in basal ganglia circuits, yielding strategies superior to standard exploration heuristics. Studies on tonic dopamine further reveal its role in biasing value learning through D1/D2 receptor dynamics (Mikhael et al., 2025). These findings suggest that artificial RL could benefit from analogous unified signalling.

Existing RL methods typically treat reward, novelty, and uncertainty as architecturally distinct components. While these signals may be combined for reward shaping, they remain separate: novelty bonuses (Burda et al., 2019) and uncertainty bonuses (Osband et al., 2016) are often computed by auxiliary networks and do not propagate into a shared learning signal. In contrast, this project proposes a dopamine-inspired framework that achieves true integration by fusing extrinsic reward (TD error), intrinsic novelty (Random Network Distillation), and epistemic uncertainty (bootstrapped ensembles) into a single biological reward prediction error (bio-RPE). Rather than applying independent bonuses, the bio-RPE unifies these signals through three neuroscientifically grounded schemes: additive, multiplicative, and gated. The magnitude of this unified signal dynamically neuromodulates the Q-network learning rate and exploration temperature, mimicking dopamine-driven synaptic plasticity. The framework is evaluated on sparse-reward benchmarks (MountainCar-v0, Acrobot-v1) and a dense-reward validation task (CartPole-v1), with multi-seed ablations isolating the contribution of each component and integration scheme.

1.1 Research Questions

This work addresses the following research questions:

- **RQ1:** How can extrinsic reward, novelty, and uncertainty be mathematically integrated into a single dopamine-inspired RPE signal (δ_{bio})?
- **RQ2:** What weighting or integration scheme (additive, multiplicative, or gating) best balances exploration and stability during training?
- **RQ3:** Does using δ_{bio} to dynamically modulate learning rate and exploration temperature improve sample efficiency and convergence stability compared to agents with fixed learning rates and exploration schedules?
- **RQ4:** How does the proposed bio-RPE framework perform across different environment types (sparse vs dense reward)?
- **RQ5:** To what extent does the agent’s learning behaviour resemble dopaminergic mechanisms observed in biological systems?

1.2 Motivation

This project is motivated by two goals. First, to improve exploration and learning stability in RL by moving beyond fixed heuristics and manually tuned intrinsic rewards. Instead of treating novelty and uncertainty as separate signals, the proposed bio-RPE integrates them into a single, adaptive learning signal to reflect the modulatory role of dopamine in biological systems. Second, the project aims to bridge the gap between neuroscience and deep RL by grounding algorithmic design choices in established neuroscience principles.

This project aims to:

- Formulate a unified bio-RPE that combines TD error, novelty, and uncertainty under ablated integration schemes.
- Incorporate this signal into a Bootstrapped DQN backbone through online neuromodulation of learning rate and exploration behaviour.
- Evaluate the resulting framework across sparse and dense environments, with multi-seed ablations isolating the contribution of each component and scheme.

2. Literature Review

This section reviews prior work relevant to this project, progressing from foundational RL theory through novelty-based exploration, uncertainty estimation, biologically inspired learning, and nonlinear signal integration. Table 2.1 summarises the key papers most relevant to this project’s design.

2.1 Reinforcement Learning and Temporal-Difference Learning

Reinforcement learning is commonly formalised using the Markov Decision Process (MDP) framework, in which an agent learns a policy that maximises expected cumulative reward through interaction with an environment (Sutton & Barto, 2018). Central to many RL algorithms is temporal-difference (TD) learning, where value estimates are updated using the TD error, defined as the difference between predicted and observed returns ($\delta_{TD} = r_t + \gamma \max_a Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)$).

The introduction of Deep Q-Networks (DQN) by Mnih et al. (2015) demonstrated that TD learning could be scaled to high-dimensional state spaces using deep neural networks, achieving human-level performance on Atari games by using experience replay to decorrelate samples and target networks to stabilise bootstrapping. Subsequent work

addressed DQN's limitations: Double DQN (Van Hasselt et al., 2016) improved data efficiency, and distributional RL (Bellemare et al., 2017) modelled full return distributions.

Despite these advances, TD-based methods depend critically on informative reward signals and often struggle in environments where rewards are sparse or delayed (Bellemare et al., 2016). MountainCar-v0's constant -1 reward and Acrobot-v1's underactuated dynamics both expose this, motivating the exploration enhancements central to this project.

2.2 Novelty-Based Exploration and Intrinsic Motivation

To combat sparse-reward environments, intrinsic motivation methods augment extrinsic rewards with auxiliary signals that encourage agents to explore novel or surprising states. The Intrinsic Curiosity Module (ICM; Pathak et al., 2017) rewards agents based on prediction error from a forward dynamics model in a learned feature space, encouraging exploration of states where dynamics are poorly understood. While effective in some settings, ICM is prone to reward hacking when the agent can generate novel observations without making task progress.

Random Network Distillation (RND; Burda et al., 2019) addresses this by measuring novelty as the mean squared error between a fixed randomly initialised target network and a trainable predictor network. Because the target is fixed, the only way to reduce prediction error is to visit states more frequently, providing a non-exploitable novelty signal. RND demonstrated strong performance in sparse-reward Atari environments such as Montezuma's Revenge.

A key limitation is rapid saturation: novelty signals decay as the agent explores, reducing guidance over time. Treating novelty as an independent reward bonus also creates a manual balancing problem between intrinsic and extrinsic objectives. The bio-RPE framework addresses both issues by integrating novelty directly into the unified error signal.

2.3 Uncertainty-Based Exploration and Ensemble Methods

Another class of exploration strategies focuses on modelling epistemic uncertainty in value estimates. Bootstrapped DQN, introduced by Osband et al. (2016), maintains an ensemble of K Q-networks trained on different subsets of experience. Exploration is driven by disagreement among ensemble members, measured as the variance of Q-value predictions, providing a principled approximation of posterior uncertainty. Randomised prior functions (Osband et al., 2018) further improved uncertainty estimation by preventing ensemble collapse through fixed prior networks added to each head's output.

Never-Give-Up (NGU; Badia et al., 2020) combined episodic novelty (embedding-based k-nearest neighbour counts) with long-term RND novelty and uncertainty to yield diverse, persistent exploration across Atari. Agent57 (Badia et al., 2020) extended this to achieve above-human performance on all 57 Atari games by meta-learning a mixture of exploratory and exploitative policies. More recently, EIPO (Liu & Swann, 2023) formalised exploration-exploitation as a constrained optimisation problem, providing theoretical guarantees on regret in sparse settings.

Despite their effectiveness, uncertainty-based methods share an important limitation relevant to this project: uncertainty is typically used only to modulate action selection, not learning rates or plasticity. The Q-network parameters are updated with a fixed learning rate regardless of agent's epistemic state. This contrasts with biological dopaminergic systems, where uncertainty and surprise directly modulate synaptic plasticity (Frémaux & Gerstner, 2016; Barry & Gerstner, 2022). The bio-RPE framework closes this gap by using ensemble-

derived uncertainty as a component of the bio-RPE, which then neuromodulates the Q-network learning rate online.

2.4 Biologically Inspired Reinforcement Learning

Neuroscience research has established that midbrain dopaminergic neurons encode an RPE signal closely related to the TD error (Schultz, Dayan, & Montague, 1997). However, subsequent research has revealed that biological dopamine signalling is considerably richer than a simple reward discrepancy. Schultz (2016, 2023), Bromberg-Martin et al. (2010), and Costa & Averbeck (2022) demonstrate that dopamine neurons respond not only to unexpected rewards, but also to novelty, uncertainty, and changes in reward contingencies. These signals dynamically modulate synaptic plasticity, effectively adjusting learning rates and behavioural flexibility over time.

Wang et al. (2024) provide direct evidence of how novelty and uncertainty interact within dopaminergic systems. Their computational model of basal ganglia circuits shows that transient dopamine responses to novelty (decaying approximately as $1/\sqrt{n}$ with familiarity) facilitate efficient uncertainty-driven exploration, yielding strategies superior to standard UCB or Thompson sampling in multi-armed bandit simulations. This finding directly motivates the inclusion of both novelty and uncertainty in the bio-RPE, and the expectation that their interaction should be multiplicative or gated rather than merely additive.

Computational models inspired by these findings include dopamine-gated Hebbian plasticity (Frémaux & Gerstner, 2016) and neuromodulated meta-reinforcement learning (Wang et al., 2018). More recently, Mikhael et al. (2025) demonstrated that tonic dopamine levels modulate baseline excitability and value biases through D1/D2 receptor dynamics, providing a mechanistic account of how dopamine concentration governs the exploration-exploitation trade-off at a cellular level. While these biological models provide compelling theoretical grounding, most remain abstract or rely on neurally detailed simulations that are difficult to integrate directly into standard deep RL frameworks, motivating the engineering approximations proposed here.

2.5 Nonlinear Signal Integration and Gating Mechanisms

The choice of how to combine the TD error, novelty, and uncertainty signals is non-trivial. Simple additive combination is the most common approach (e.g., Burda et al., 2019), but it treats the signals as independent and cannot capture interactions between them.

Multiplicative interactions have been studied in the context of recurrent neural networks: Multiplicative Integration RNNs (MI-RNNs; Wu et al., 2016) employ Hadamard (element-wise) products to gate information flow, yielding more stable gradient propagation than additive alternatives.

From a biological perspective, prefrontal-striatal pathways implement a cognitive control gate that switches dopaminergic systems between exploratory and exploitative modes depending on context (Frank et al., 2007). Yu & Dayan (2005) also demonstrated that novelty and uncertainty drive exploration through partially antagonistic neuromodulatory pathways, with norepinephrine signalling unexpected uncertainty (akin to novelty) and acetylcholine signalling expected uncertainty. These mechanisms collectively motivate the gated integration scheme proposed in this project.

In the RL literature, gated reward accumulation (G-RA; Zhao et al., 2025) applies multiplicative gating to combine sparse long-horizon extrinsic rewards with critic estimates, demonstrating improved credit assignment in delayed-reward tasks. Multiplicative couplings have also been shown to accelerate learning in biologically inspired networks, where gating

replicates neuromodulatory control of synaptic gain (Zhang et al., 2025). These precedents motivate the multiplicative and gated schemes evaluated in this project.

2.6 Hybrid and Multi-Signal Approaches

Several recent approaches attempt to combine multiple exploration signals, providing the closest points of comparison for this project. Xu et al. (2021) propose the SurNor model, which explicitly separates novelty (state unfamiliarity) from surprise (reward prediction error) and uses each signal for a distinct purpose: novelty drives exploration of unfamiliar states, while surprise modulates the learning rate when environmental statistics change. SurNor accounts for patterns in human exploratory behaviour and neural data, demonstrating the value of combining multiple signals within a coherent framework. However, SurNor operates in tabular or small discrete spaces and does not scale to deep RL with continuous or high-dimensional state spaces.

The Hybrid Intrinsic Reward for Exploration (HIRE; Zhu et al., 2025) combines count-based (Ostrovski, 2017), RND, and curiosity bonuses via a learned weighting mechanism, demonstrating that multi-signal intrinsic rewards improve exploration across diverse tasks. However, HIRE is not biologically grounded and does not incorporate uncertainty-driven neuromodulation of learning dynamics. Kaplanis et al. (2024) show that dynamic learning rate control substantially improves continual learning, but their signal does not incorporate novelty or uncertainty.

Go-Explore (Ecoffet et al., 2021) and EIPO (Liu & Swann, 2023) represent strong baselines for hard sparse-reward problems. Go-Explore achieves state-of-the-art results on Montezuma’s Revenge by explicitly storing and returning to promising states, but requires domain-specific state representations. EIPO offers principled regret bounds but is difficult to combine with function approximation. The bio-RPE framework, by contrast, integrates naturally with standard deep RL architectures at minimal computational overhead.

2.7 Summary of Key Related Work

Table 2.1 summarises the papers most directly related to this project.

Table 2.1: Comparison of key related works. ✓ = incorporated, ✗ = not incorporated, Partial = partially incorporated (e.g. SurNoR uses surprise to modulate learning rate but not as a full uncertainty estimate). ‘Neuromod. LR’ = dynamic learning rate modulation based on signal magnitude.

Paper	TD error	Novelty	Uncertainty	Neuromod. LR	Scheme Ablation
Burda et al. (2019) — RND	✓	✓	✗	✗	✗
Osband et al. (2016) — Boot. DQN	✓	✗	✓	✗	✗
Xu et al. (2021) — SurNoR	✓	✓	Partial	✓	✗
Badia et al. (2020) — NGU	✓	✓	✓	✗	✗
Zhu et al. (2025) — HIRE	✓	✓	✗	✗	✗
Wang et al. (2024)	✓	✓	✓	✗	✗
Kaplanis et al. (2024)	✓	✗	✗	✓	✗
This project — bio-RPE	✓	✓	✓	✓	✓

2.8 Research Gap and Project Positioning

As Table 2.1 makes clear, no existing framework unifies TD learning, novelty-based exploration, uncertainty estimation, and biological grounding within a single ablated deep RL pipeline that neuromodulates learning dynamics. SurNoR and Wang et al. (2024) came closest biologically but operate in simplified settings. NGU and HIRE combine multiple intrinsic signals but lack biological motivation and do not modulate learning rates. Bootstrapped DQN provides principled uncertainty but uses it only for action selection, not plasticity. No prior work has systematically ablated integration scheme choice (additive vs. multiplicative vs. gated) within this context.

This project fills the gap by proposing a biologically grounded bio-RPE that: (i) fuses TD error, novelty, and uncertainty into a single composite signal under three rigorously ablated integration schemes; (ii) uses this signal to simultaneously neuromodulate learning rate and exploration temperature; and (iii) evaluates the resulting system across sparse-reward (MountainCar-v0, Acrobot-v1) and dense-reward (CartPole-v1) environments with multi-seed ablations. The integration of novelty (RND) and uncertainty (ensemble variance) into the error signal, rather than as independent reward bonuses, is what distinguishes this framework from prior work and is its primary algorithmic novelty.

3. Methodology

This section describes the design and implementation of the dopamine-inspired reinforcement learning framework proposed in this project. It is organised in five parts: the baseline architecture, novelty and uncertainty estimation modules, the bio-RPE formulation and its three integration modes, the neuromodulation mechanism, and the experimental setup. All experiments were conducted on Google Cloud Platform using an Ubuntu 22.04 LTS virtual machine (e2-standard-4; 4 vCPUs, 16 GB RAM, 50 GB persistent storage). Python 3 with PyTorch (Paszke et al., 2019), Gymnasium (Towers et al., 2023), NumPy, Pandas, Scipy, and Matplotlib was used for implementation, training, logging, and analysis.

3.1 Algorithmic Design

3.1.1 Baseline Architecture: Bootstrapped Deep Q-Network

The core learning algorithm is based on Bootstrapped Deep Q-Network (Bootstrapped DQN), proposed by Osband et al. (2016), extended here with Double DQN action selection (Van Hasselt et al., 2016).

Bootstrapped DQN is selected as the backbone for two reasons directly relevant to this project. First, its ensemble of K Q-networks enables deep exploration through Thompson sampling-style head selection. Second, and critically, ensemble disagreement provides a principled approximation of epistemic uncertainty without requiring a separate auxiliary network, making it a natural and computationally efficient source for the uncertainty component of the bio-RPE.

Double DQN action selection (Van Hasselt et al., 2016) decouples action selection from its evaluation, preventing the overestimation bias of standard Q-learning and ensuring that the baseline δ_{TD} remains a stable reference point before being fused with intrinsic novelty and uncertainty.

Implementation Details:

- An ensemble of $K = 5$ Q-networks is maintained, each parameterised by a multilayer perceptron with two hidden layers of 128 units and ReLU activations. A

corresponding ensemble of 5 target networks is maintained separately and used exclusively for computing Bellman targets, to prevent overestimation bias.

- Each episode begins by selecting a single ensemble head uniformly at random to act as the behavioural policy. Action selection follows an ϵ -greedy policy initialised at $\epsilon = 1.0$, subsequently decayed by the neuromodulation mechanism (Section 3.4).
- Transitions $(s, a, r, s', done)$ are stored in a 20,000-capacity shared replay buffer. A training step is executed every 4 environment steps, sampling a batch of 64 transitions uniformly at random
- A Bernoulli-distributed bootstrapped mask ($p = 0.8$) is applied to each sample's TD loss per head, promoting ensemble diversity and improving uncertainty estimation.

Within each training step and for each head K , the TD target is computed using Double DQN: the online network selects the greedy action in the next state, and the target network evaluates its value. Formally, for a transition $(s, a, r, s', done)$:

$$y_k = r + \gamma \cdot Q_{target,k} \left(s', \operatorname{argmax}_{a'} Q_{online,k}(s', a') \right) \cdot (1 - done) \quad (1)$$

The objective is to minimize the masked mean squared TD error:

$$L_k = \frac{\sum_i [mask_i \cdot (y_{k,i} - Q_{online,k}(s_i, a_i))^2]}{\sum_i mask_i + \epsilon} \quad (2)$$

where the small constant $\epsilon = 10^{-6}$ prevents division by zero. Each head's parameters are updated independently using the Adam optimiser (Kingma & Ba, 2015; LR: 3×10^{-4} ; gradient clipping at a maximum of L2 norm of 10.0). Finally, target networks are updated by hard copying the online networks' weights at the end of each episode.

3.1.2 Novelty Estimation via Random Network Distillation

Novelty is estimated using Random Network Distillation (RND), introduced by Burda et al. (2019). RND consists of: (i) a fixed, randomly initialised target network f_{target} , and (ii) a trainable predictor network f_{pred} , trained to predict the target network's outputs, where θ is the parameters being learned.

For a given state s , the novelty signal δ_{nov} is defined as the mean squared error between the predictor and target outputs:

$$\delta_{nov} = \|f_{pred}(s; \theta) - f_{target}(s)\|^2 \quad (3)$$

Prediction error is high in unfamiliar states and decays with repeated exposure as the predictor converges, mirroring the habituation of novelty responses in biological dopaminergic neurons (Ljungberg et al., 1992; Schultz, 2016; Wang et al., 2024). The RND predictor is updated online every training step using the same replay buffer batches, with the Adam optimiser (LR = 3×10^{-4} , selected through preliminary experimental tests). For ablation conditions without novelty (TD, TD+UNC), the use_novelty flag replaces δ_{nov} with a zero tensor, removing its contribution to the bio-RPE.

3.1.3 Uncertainty Estimation via Ensemble Disagreement

Epistemic uncertainty, δ_{unc} , is estimated from the disagreement (or "variance") among the K Q-network ensemble heads. Q-values from all heads are stacked into a tensor of shape $[K, batch_size, action_dim]$; variance across heads is computed per state-action pair and averaged across actions:

$$\delta_{unc}(s) = \frac{1}{|A|} \sum_{a \in A} \left(\frac{1}{K} \sum_{k=1}^K (Q_k(s, a) - \bar{Q}(s, a))^2 \right) \quad (4)$$

where:

- K : Total number of heads in the ensemble ($K=5$)
- $|A|$: The size of the action space
- $Q_k(s, a)$: The Q-value predicted by the k -th head for state s and action a
- $\bar{Q}(s, a)$: The ensemble mean, defined as $\frac{1}{K} \sum_{k=1}^K Q_k(s, a)$.

This quantity is highest in sparse or inconsistent regions of the state space, providing a Bayesian approximation of posterior uncertainty over value estimates (Osband et al., 2016). To maintain numerical stability, estimates are z-score normalised within each batch, clipped at 0 via ReLU to remove negative values, and clamped at a maximum of 5.0. Conditions without uncertainty (TD, TD+NOV) replace δ_{unc} with a zero tensor via the use_uncertainty flag.

The complete training procedure is formalised in Algorithm 1:

Algorithm 1: Bootstrapped DQN Training with bio-RPE
Input: env, episodes E, replay buffer B, ensemble size K, integration mode
Initialise $\{Q_k\}^K$, $\{Q^{\wedge}_k\}^K$, f_pred, f_targ (frozen), $B \leftarrow \emptyset$
for episode = 1 to E do
 $s \leftarrow \text{env.reset}()$; select active head $k \sim \text{Uniform}\{1, \dots, K\}$
 for each timestep t do
 select a via ϵ -greedy under $Q_k(s, \cdot)$
 execute a; observe r, s', done; store (s, a, r, s', done)
in B; $s \leftarrow s'$
 if t mod 4 = 0 and $|B| \geq \text{batch_size}$ then
 sample minibatch from B
 compute δ_{bio} , LR(t), $\epsilon(t)$ via Algorithm 2
 update $\{Q_k\}$ with bootstrapped masking using LR(t)
 update f_pred via MSE(f_pred(s), f_targ(s)) [if novelty active]
 end if
 if done then break
 end for
 hard-copy $\{Q^{\wedge}_k\} \leftarrow \{Q_k\}$ [end of episode]
end for

Algorithm 1. Outer training loop for the bio-RPE framework. The inner bio-RPE computation and neuromodulation procedure is detailed in Algorithm 2 (Section 3.4).

3.2 Dopamine-Inspired Reward Prediction Error

The central contribution of this framework is the bio-RPE (δ_{bio}): a unified prediction error fusing TD error, novelty, and uncertainty into a single signal, analogous to the multidimensional RPE emitted by midbrain dopaminergic neurons (Schultz et al., 1997; Bromberg-Martin et al., 2010). Three integration schemes are implemented, each grounded in a distinct neuroscientific principle and corresponding to a different hypothesis about how these informational dimensions should interact.

3.2.1 Additive Integration – Linear Synaptic Summation

The additive mode is used by the TD, TD+NOV, TD+UNC, and FULL conditions. The bio-RPE is computed as a weighted sum of the three component signals:

$$\delta_{\text{bio}} = \delta_{\text{TD}} + \alpha(t) \cdot \delta_{\text{nov}} + \beta(t) \cdot \delta_{\text{unc}} \quad (5)$$

where:

- δ_{TD} is the standard temporal-difference error,
- δ_{nov} is the novelty signal from RND,
- δ_{unc} is the uncertainty signal from ensemble disagreement,
- $\alpha(t)$ and $\beta(t)$ are adaptive weighting coefficients (updated at each training step).

This scheme is grounded in linear synaptic summation: a postsynaptic dopaminergic neuron performs a weighted sum of inputs from distinct afferent circuits (novelty signals from hippocampus-VTA pathways; uncertainty from Prefrontal Cortex) producing a composite RPE comparable to the McCulloch-Pitts formulation $\sum(w_i \cdot x_i)$ (Bromberg-Martin et al., 2010; Kakade & Dayan, 2002). The adaptive coefficients mirror phasic dopamine's gain modulation of striatal responses: larger TD errors drive stronger weighting of intrinsic signals, amplifying novelty and uncertainty contributions when prediction errors are high.

3.2.2 Multiplicative Integration – Gain modulation and Precision-Weighted Errors

The multiplicative mode is used exclusively by the MULT condition. Novelty and uncertainty scale the TD error rather than adding to it:

$$\delta_{bio} = \delta_{TD} \cdot (1 + \alpha \cdot \tilde{n}_{nov} + \beta \cdot \tilde{u}_{unc}) \quad (6)$$

where $\tilde{n}_{nov}$ and $\tilde{u}_{unc}$ are novelty and uncertainty normalised to the batch mean and clamped to $[0, 1]$:

$$\tilde{n}_{nov} = clamp\left(\frac{\delta_{nov}}{mean(\delta_{nov})} + \epsilon, 0, 1\right) \quad (7)$$

$$\tilde{u}_{unc} = clamp\left(\frac{\delta_{unc}}{mean(\delta_{unc})} + \epsilon, 0, 1\right) \quad (8)$$

This scheme is grounded in two neuroscientific principles. First, neuromodulators act as gain controls. They multiply the effective weight of prediction errors depending on context, matching scaled prediction-error models in basal ganglia circuitry where uncertainty scales the dopamine-like RPE (Doya, 2002). Second, Bayesian brain theories propose precision-weighted prediction errors where the brain scales each error by the estimated precision of the signal, and so high-novelty and high-uncertainty states drive larger updates (Friston, 2009). When both n_{nov} and u_{unc} are low (familiar, well-understood states), the scale factor approaches 1.0 and the condition reduces to standard TD learning, enabling a natural transition from exploration- to exploitation-dominated behaviour as training progresses.

3.2.3 Gated Integration – Dynamic Resource Allocation and Epistemic Transition

The gated mode is used exclusively by the GATE condition. A sigmoid gate computed from the uncertainty signal allocates integration weight between novelty and uncertainty:

$$gate(s) = \sigma(k \cdot (\delta_{unc}(s) - \tau)) \quad (9)$$

$$\delta_{bio} = \delta_{TD} + \alpha \cdot (1 - gate) \cdot \delta_{nov} + \beta \cdot gate \cdot \delta_{unc} \quad (10)$$

where σ is the sigmoid function, $k = 5.0$ controls the transition steepness and $\tau = 0.5$ sets the uncertainty threshold. This scheme is grounded in three neuroscientific mechanisms: (i) prefrontal-striatal circuits implement cognitive control gates that selectively activate processing pathways depending on context, with dopaminergic systems switching between exploration and exploitation modes (Frank et al., 2007); (ii) the information-seeking value encoded from the Prefrontal Cortex determines when an organism switches from broad novelty search to focused uncertainty refinement (Gottlieb & Oudeyer, 2013; Gottlieb et al., 2014), approximated here by the gate threshold τ ; and (iii) novelty and uncertainty may

operate through parallel, partially antagonistic pathways with top-down control selecting the dominant channel (Yu & Dayan, 2005).

When uncertainty is high ($\delta_{unc} \gg \tau$), the gate approaches 1 and δ_{bio} is dominated by the uncertainty signal; when uncertainty is low ($\delta_{unc} \ll \tau$), the gate approaches 0 and the novelty signal dominates. The constants k and τ are treated as fixed hyperparameters representing the agent’s “epistemic sensitivity”.

Following all three integration modes, the bio-RPE is symmetrically clamped to the range $[-50, 50]$ to prevent numerically extreme values from destabilising the neuromodulation functions.

3.3 Neuromodulation of Learning Rates and Exploration

3.3.1 Adaptive weighting coefficients

The coefficients $\alpha(t)$ and $\beta(t)$ that scale novelty and uncertainty in the additive and gated bio-RPE are updated from the TD error via a tanh transformation:

$$signal = \tanh\left(\frac{|\delta_{TD}|}{10.0}\right) \quad (11)$$

$$\alpha(t) = \alpha_{max} \cdot signal, \quad \beta(t) = \beta_{max} \cdot signal \quad (12)$$

where $\alpha_{max} = \beta_{max} = 0.05$, which were selected empirically after lower values (0.01) provided insufficient signals, while higher values (0.1-1.0) induced training instability. This ensures that α and β are bounded within $[0, \alpha_{max}]$ and scale proportionally with the prediction error magnitude. This means large TD errors (indicating unexpected outcomes) elicit stronger weighting of novelty and uncertainty, while small TD errors (indicating accurate predictions) reduce the intrinsic signal contribution. The division by 10.0 sets the saturation point of tanh at approximately $|\delta_{TD}| = 10$, calibrated to the typical range of TD errors observed during early training.

Biologically, this mirrors phasic dopamine’s gain modulation: stronger dopamine bursts transiently increase striatal synaptic gain, amplifying the weight of concurrent novelty and uncertainty inputs (Bromberg-Martin et al., 2010).

3.3.2 Learning rate modulation

The magnitude of the bio-RPE is used to dynamically adjust the learning rate, implementing a computational analogue of dopaminergic neuromodulation of synaptic plasticity and behavioural flexibility (Frémaux & Gerstner, 2016; Cools, 2008). The learning rate of each Q-network head’s Adam optimiser is scaled at each training step by a factor derived from the bio-RPE magnitude:

$$LR_{scale} = clip\left(0.5 + 0.5 \cdot \tanh\left(\frac{|\delta_{bio}|}{10.0}\right), 0.5, 1.0\right) \quad (13)$$

$$LR(t) = LR_{base} \cdot LR_{scale} \quad (14)$$

where $LR_{base} = 3 \times 10^{-4}$. The learning rate is therefore bounded within $[\frac{LR_{base}}{2}, LR_{base}]$, increasing toward its base value during periods of high bio-RPE (active learning) and decreasing toward half its base value during periods of low bio-RPE (convergence). This mimics the dopaminergic modulation of synaptic plasticity, wherein high dopamine activity facilitates stronger learning from recent experience, while low dopamine protects established weights (Frémaux & Gerstner, 2016).

3.3.3 Exploration modulation

The exploration parameter ε is updated at each training step using a target value derived from the TD error signal, smoothed by exponential moving average:

$$\varepsilon_{target} = \varepsilon_{min} + (\varepsilon_{base} - \varepsilon_{min}) \cdot signal \quad (15)$$

$$\varepsilon(t) = (1 - \varepsilon_{smooth}) \cdot \varepsilon(t-1) + \varepsilon_{smooth} \cdot \varepsilon_{target} \quad (16)$$

$$\varepsilon(t) = clip(\varepsilon(t), \varepsilon_{min}, 1.0) \quad (17)$$

where $\varepsilon_{min} = 0.01$, $\varepsilon_{base} = 1$, $\varepsilon_{smooth} = 0.01$, and $signal = \tanh\left(\frac{|\delta_{bio}|}{10.0}\right)$ as defined above. Large prediction errors drive ε toward higher exploration; small errors drive ε toward ε_{min} , favouring exploitation. The smoothing factor ε_{smooth} ensures that epsilon changes gradually rather than responding to individual noisy observations. This reproduces tonic dopamine regulation of behavioural flexibility: elevated tonic dopamine promotes exploration while reduced levels favour habitual responding (Cools, 2008; Humphries et al., 2012).

3.4 Framework Overview

Algorithm 2 formalises the bio-RPE computation across all three integration modes. Figure 3.1 illustrates the full pipeline from signal extraction to neuromodulatory output.

Algorithm 2: bio-RPE Computation and Neuromodulation

Input: minibatch $(s, a, r, s', done)$, $\{Q_k\}$, $\{Q^*_k\}$, f_{pred} , f_{targ} , mode, α_{max} , β_{max}

```

// Signal extraction
 $\delta\_TD \leftarrow \text{mean}_k [ r + \gamma \cdot \max_{a'} Q_k(s', a') - Q_k(s, a) ]$ 
// Double DQN target
 $\delta\_nov \leftarrow \text{MSE}(f_{pred}(s), f_{targ}(s))$ 
// RND novelty
 $\delta\_unc \leftarrow \text{mean}_a \text{Var}_k [ Q_k(s, a) ]$ 
// ensemble variance

// Adaptive weighting
 $\alpha(t) \leftarrow \alpha_{max} \cdot \tanh(|\delta\_TD| / 10);$ 
 $\beta(t) \leftarrow \beta_{max} \cdot \tanh(|\delta\_TD| / 10)$ 

// Integration scheme
if mode = ADD then
     $\delta\_bio \leftarrow \delta\_TD + \alpha(t) \cdot \delta\_nov + \beta(t) \cdot \delta\_unc$ 
else if mode = MULT then
     $\delta\_bio \leftarrow \delta\_TD \cdot (1 + \alpha(t) \cdot \delta\_nov + \beta(t) \cdot \delta\_unc)$ 
else if mode = GATE then
    gate  $\leftarrow \sigma(\delta\_unc - \tau\_unc)$ 
     $\delta\_bio \leftarrow \delta\_TD + \alpha(t) \cdot (1 - \text{gate}) \cdot \delta\_nov + \beta(t) \cdot \text{gate} \cdot \delta\_unc$ 
end if
 $\delta\_bio \leftarrow \text{clamp}(\delta\_bio, -50, 50)$ 

// Neuromodulation
LR_scale  $\leftarrow \text{clip}(0.5 + 0.5 \cdot \tanh(|\delta\_bio|/10), 0.5, 1.0)$ 
LR(t)  $\leftarrow \text{LR\_scale} \cdot \text{LR\_base}$ 
 $\varepsilon_{target} \leftarrow \varepsilon_{min} + (1 - \varepsilon_{min}) \cdot \tanh(|\delta\_bio|/10)$ 
 $\varepsilon(t) \leftarrow (1 - \varepsilon_{smooth}) \cdot \varepsilon(t-1) + \varepsilon_{smooth} \cdot \varepsilon_{target}$ 

Return  $\delta\_bio, LR(t), \varepsilon(t)$ 

```

Algorithm 2. bio-RPE computation and neuromodulation for a single training step. σ denotes the sigmoid function; clamp restricts δ_{bio} to $[-50, 50]$.

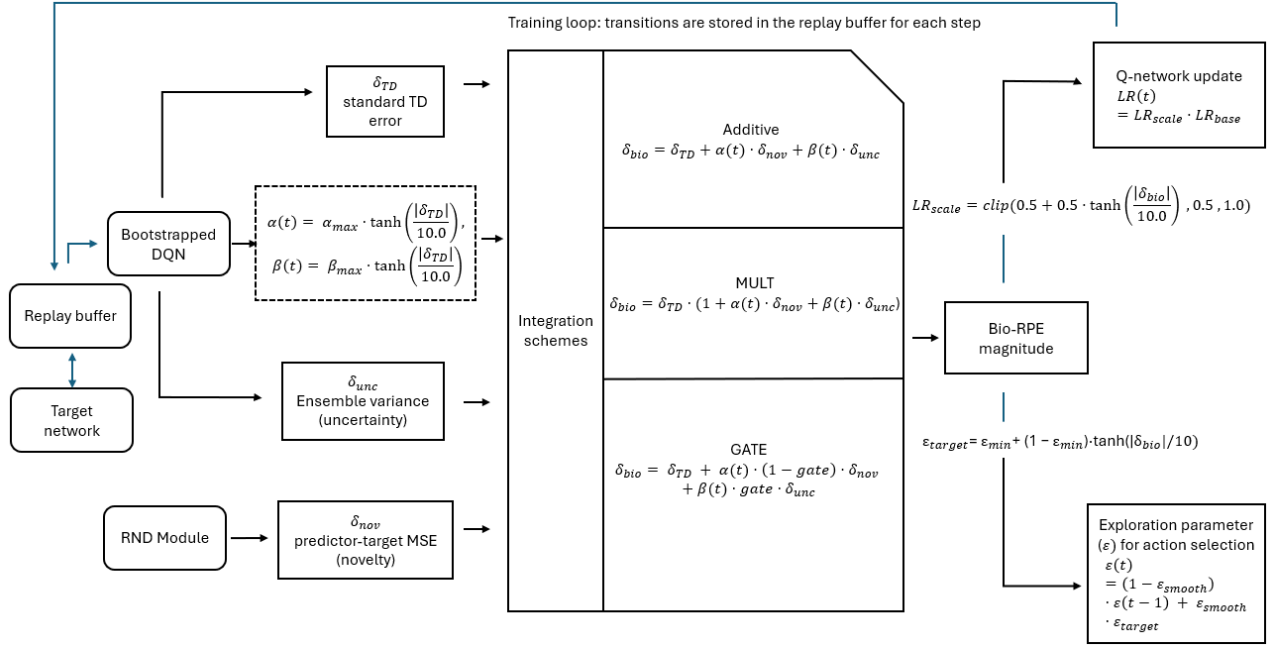


Figure 3.1. Architecture of the bio-RPE framework. The Bootstrapped DQN produces the standard TD error (δ_{TD}) and ensemble variance (δ_{unc}); the RND module produces the novelty signal (δ_{nov}). Adaptive weighting coefficients $\alpha(t)$ and $\beta(t)$ are derived from the TD error magnitude via a tanh transformation. One of three integration schemes — Additive, Multiplicative (MULT), or Gated (GATE) — combines these signals into the bio-RPE (δ_{bio}). The bio-RPE magnitude then drives two neuromodulatory outputs: the learning rate scale $LR(t)$ applied to each Q-network head, and the exploration temperature $\epsilon(t)$ governing action selection. Transitions are stored in the replay buffer at each step (top loop).

3.5 Environments

Experiments were conducted on three environments from the OpenAI Gymnasium library (Brockman et al., 2016), selected to span a range of exploration difficulty and reward density. The key properties of each environment are summarised in Table 3.2.

Table 3.2. Environment specifications. State dimensions, action spaces, reward structures, and training episode budgets are shown for each environment.

Environment	State dim.	Actions	Reward	Episodes	Eval every
CartPole-v1	4	2 (discrete)	+1 per step	500	25 episodes (5 greedy)
MountainCar-v0	2	3 (discrete)	-1 per step	700	25 episodes (5 greedy)
Acrobot-v1	6	3 (discrete)	-1 per step	500	25 episodes (5 greedy)

3.5.1 CartPole-v1

CartPole-v1 is a dense-reward control task in which a pole is balanced on a cart by applying discrete left or right forces. The state space is four-dimensional, comprising cart position, cart velocity, pole angle, and pole angular velocity. A reward of +1 is issued at every timestep the pole remains within bounds; episodes terminate when the pole exceeds 15 degrees from vertical, the cart leaves its fixed range, or 500 timesteps elapse (maximum return: 500). This environment serves as a validation task, confirming that the bio-RPE framework does not impede learning under a well-shaped reward signal.

3.5.2 MountainCar-v0

MountainCar-v0 is a sparse-reward continuous control task in which an underpowered car must build momentum through oscillation to reach a goal position on a hill. The two-dimensional state space encodes position and velocity. A reward of -1 is issued at every

timestep until the goal is reached or the 200-step episode limit is hit, giving a minimum per-episode return of -200. This environment is the primary testbed for evaluating exploration strategies, as the goal region cannot be reached through random or minimally guided exploration within a short time horizon, making it a well-established benchmark for intrinsic motivation methods. The episode budget is 700 to allow adequate time for all conditions to reach the goal region.

3.5.3 Acrobot-v1

Acrobot-v1 is a moderately sparse-reward task in which a two-link pendulum must be swung above a threshold height. The six-dimensional state space encodes the sine and cosine of both joint angles and both joint angular velocities. A reward of -1 is issued per timestep until the goal height is achieved, with a maximum episode length of 500 steps. Acrobot was included to assess whether the benefits of bio-RPE augmentation generalise beyond MountainCar’s specific dynamics.

3.6 Experimental Conditions

Six experimental conditions were evaluated in a controlled ablation design. Each condition corresponds to a specific configuration of the `use_novelty` and `use_uncertainty` flags and the `integration_mode` parameter, enabling systematic isolation of each component’s contribution. The conditions are summarised in Table 3.1.

Table 3.1. *Experimental conditions evaluated in this study. The TD condition serves as the baseline; all other conditions introduce one or more components of the bio-RPE framework.*

Condition	δ_{nov}	δ_{unc}	Mode	Description
TD	X	X	-	Standard bootstrapped DQN baseline. No intrinsic signals
TD+NOV	✓	X	Additive	Novelty-only augmentation via RND. Tests the isolated effect of curiosity-driven exploration.
TD+UNC	X	✓	Additive	Uncertainty-only augmentation via ensemble disagreement. Tests the isolated effect of epistemic uncertainty.
FULL	✓	✓	Additive	Both novelty and uncertainty combined additively. Full bio-RPE signal.
MULT	✓	✓	Multiplicative	Both signals integrated multiplicatively. TD error is scaled by a novelty and uncertainty dependent factor.
GATED	✓	✓	Gated	Sigmoid gate allocates weight between novelty and uncertainty

Conditions sharing the same integration mode but differing in active signals (e.g., TD+NOV vs. TD+UNC) attribute performance differences to individual components. Conditions differing only in integration mode (FULL vs. MULT vs. GATE) compare the three fusion strategies directly.

3.7 Training Protocol

3.7.1 Initialisation and reproducibility

All experiments used ten random seeds (0-9). For each seed, Python, NumPy, and PyTorch RNGs were seeded identically via `set_seed()`, with each episode’s environment reset seeded as `seed + episode_number` to ensure varied but reproducible initial states.

3.7.2 Training Loop

Within each episode, the agent interacts with the environment in a standard RL loop. At the start of each episode, one ensemble head is randomly selected via `begin_episode()`.

At each timestep, the agent selects an action via ε -greedy policy, observes the transition, stores it in the replay buffer, and logs the state in visited_states log. Every 4 timesteps, a training step is executed: a batch of 64 transitions updates all ensemble heads (with bootstrapped masking), the RND predictor (if active), and neuromodulation parameters. Target networks are hard-copied from online networks at each episode end. The full procedure is formalised in Algorithm 1.

3.7.3 Evaluation protocol

Greedy evaluation is conducted every 25 training episodes. ε is temporarily set to 0.0, the environment is reset, and the agent runs for 5 complete greedy episodes. The mean total reward across these 5 episodes is recorded as the evaluation score for that checkpoint. After evaluation, epsilon is restored to its current training value. This decoupled evaluation protocol ensures that exploration noise does not contaminate performance estimates.

3.8 Evaluation Metrics and Output

Table 3.3 details the metrics recorded throughout each run.

Table 3.3 Recorded Metrics.

Metric	Description
Episodic training reward	Total undiscounted reward per training episode under the current exploration policy. Reflects the agent’s performance.
Greedy evaluation reward	Mean reward across 5 greedy evaluation episodes, recorded every 25 training episodes.
TD error ($ \delta_{TD} $)	Absolute mean TD error across the batch, averaged over ensemble heads.
Novelty signal (δ_{nov})	Mean RND prediction error across the batch. Zero for conditions without novelty.
Uncertainty signal (δ_{unc}):	Mean normalised ensemble variance across the batch. Zero for conditions without uncertainty.
Bio-RPE magnitude ($ \delta_{bio} $):	Absolute mean bio-RPE after clamping.
Exploration epsilon (ε):	Exploration probability at episode end, reflecting the neuromodulation state.
Novelty and uncertainty weighting coefficients ($\alpha(t)$ and $\beta(t)$)	Strength of novelty ($\alpha(t)$) and uncertainty $\beta(t)$
Learning rate (LR)	The aggressiveness of the bio-RPE driven optimizer
State visitation heatmaps:	2D histograms of visited states across all seeds, projected onto two interpretable dimensions per environment (e.g., position/velocity for MountainCar; joint angles for Acrobot).

All metrics except state visitation heatmaps are reported as mean $\pm$ standard deviation across ten seeds. Per-seed raw curves are also saved for reference. Summary statistics include the final and best evaluation reward per seed with means and standard deviations computed across seeds.

3.9 Hyperparameter Summary

All hyperparameters were held constant across all conditions and environments to ensure fair comparison. This may disadvantage the MULT and GATE conditions relative to their tuned potential, which is discussed as a limitation in Section 5.

Table 3.4. Complete hyperparameter configuration used in all experiments

Hyperparameter	Value	Description
Ensemble size (K)	5	Number of Q-network heads in the bootstrap ensemble
Learning Rate (LR)	3×10^{-4}	Base Adam optimizer learning rate; dynamically scaled by neuromodulation
Discount factor (γ)	0.99	Future reward discount for Bellman targets
Replay buffer size	20,000	Maximum number of transitions sampled per training step
Batch size	64	Number of transitions sampled per training step
Bootstrap mask prob.	0.8	Bernoulli probability that each sample contributes to each head’s update
Gradient clip norm	10.0	Maximum L2 norm for gradient clipping per Q-network head
α_{max}	0.05	Maximum novelty weighting coefficient in bio-RPE
β_{max}	0.05	Maximum uncertainty weighting coefficient in bio-RPE
ϵ_{min}	0.01	Minimum exploration probability (floor for ϵ -greedy policy)
ϵ_{smooth}	0.01	Exponential smoothing factor for epsilon update
Target update	Hard (per episode)	Target networks fully copied from online networks after each episode
Train frequency	Every 4 steps	One training is performed per 4 environment interactions
RND hidden dim	128	Hidden layer size for both RND target and predictor networks
RND learning rate	3×10^{-4}	Adam learning rate for the RND predictor network
Gate threshold (τ)	0.5	Uncertainty threshold for sigmoid gate in GATE integration mode
Gate steepness (k)	5.0	Sigmoid slope parameter for GATE integration mode

4. Results

4.1 Experimental Overview

This section presents the empirical results of the dopamine-inspired reinforcement learning framework evaluated across the three OpenAI Gymnasium environments (CartPole-v1, MountainCar-v0, and Acrobot-v1) under six experimental conditions. All experiments were conducted with ten random seeds per condition per environment, representing a total of 180 individual training runs ($10 \text{ seeds} \times 6 \text{ conditions} \times 3 \text{ environments}$). All six conditions completed training without numerical instability (no NaN values or gradient explosions) across all 180 runs, confirming that the bio-RPE formulation is computationally tractable. However, outcome variance across seeds was substantial in some conditions, with stuck seeds and crashed runs documented in Table 4.6, reflecting sensitivity to weight initialisation rather than numerical failure. Table 3.1 (Section 3.6) summarises the conditions evaluated.

Performance is assessed along two primary dimensions: final evaluation reward (greedy evaluation at the last training checkpoint) and best evaluation reward (the highest greedy evaluation score achieved across all checkpoints). Following Agarwal et al. (2021), median and interquartile range (IQR) are reported alongside mean $\pm$ standard deviation as alternatives in the presence of outlier seeds. For MountainCar-v0 and Acrobot-v1, reward is inherently negative and higher values indicate better performance. All statistics are reported

across ten seeds. Evaluation was conducted every 25 episodes using five greedy episodes per checkpoint ($\varepsilon = 0$).

4.1.1 Statistical Testing Methodology

To assess whether observed performance differences are statistically reliable, Mann-Whitney U tests (Wilcoxon rank-sum) are applied pairwise between each augmented condition and the TD baseline, separately for each environment. This non-parametric test is appropriate given the small sample size ($n=10$ seeds) and non-normal reward distributions produced by stuck and crashed seeds. A Bonferroni correction is applied to control for multiple comparisons: with 5 conditions $\times$ 3 environments = 15 tests, the corrected significance threshold is $\alpha = 0.05/15 = 0.0033$. Results at uncorrected $p < 0.05$ are reported as trending and denoted (*); results surviving Bonferroni correction are denoted (**). It is acknowledged that with only 10 seeds and the observed inter-condition variance, most pairwise differences are unlikely to survive correction. This is reported as a null finding rather than a failure, consistent with the limitations discussed in Section 5.6.

4.2 CartPole-v1

4.2.1 Overview and Training Dynamics

CartPole-v1 is the dense-reward validation task (see Section 3.5.1). All six conditions demonstrate clear and sustained learning curves across ten seeds, confirming that the bio-RPE framework does not impede learning when extrinsic rewards are frequent and well-shaped.

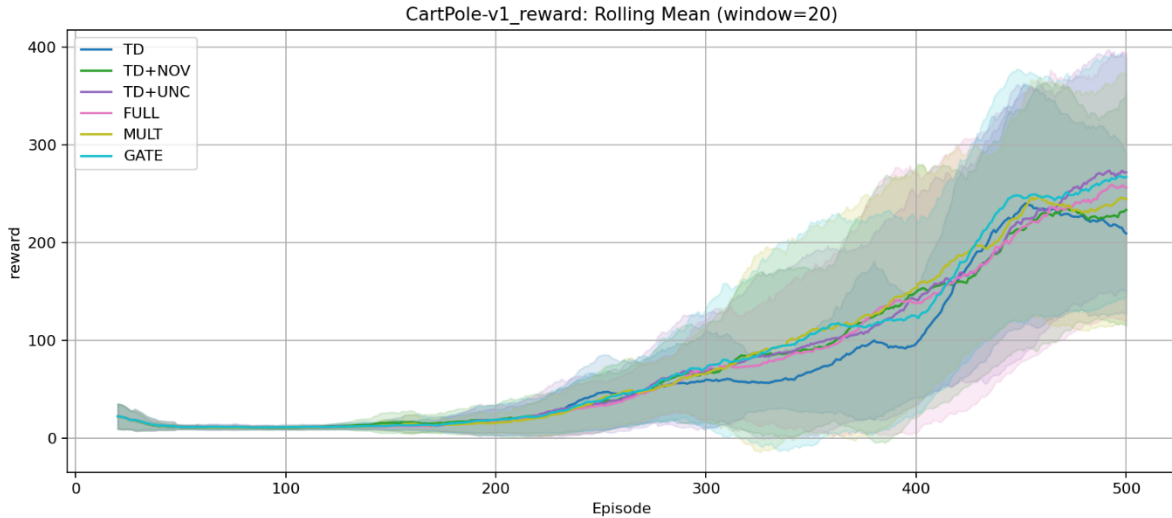


Figure 4.1. CartPole-v1 training reward (rolling mean, window = 20 episodes, mean $\pm$ std over 10 seeds). Curves begin at Episode 20, when the first full rolling average becomes available.

All conditions remain flat near the random-policy baseline (~ 9 -20 reward units) for the first 100-200 episodes, after which a rapid performance inflection is observed between episodes 250-300, rising from ~ 20 to ~ 70 reward units, as value estimates stabilise and the ε -greedy policy begins to exploit learned knowledge effectively. In episodes 300-500, performance diverges: TD+UNC, MULT and FULL exhibit steeper late-training gains (reaching above 250 units at episode 500), while TD lags behind at ~ 210 .

4.2.2 Evaluation Performance

Table 4.2 presents the final and best evaluation reward statistics, including median and IQR. TD+UNC achieves the highest final evaluation mean (320.3 ± 122.9), best mean (412.6) and the highest median (348.1), reflecting that its strongest seeds reach near-maximum performance. It is followed by GATE (276.2 ± 85.7 , median 273.5) and TD+NOV (275.4 ± 115.0 , median 269.5). The TD baseline ranks last on final evaluation (215.4 ± 86.1 , median 207.9), indicating that bio-augmented conditions do provide a learning advantage in dense-reward settings. The evaluation curves in Figure 4.2 confirm that all conditions track closely until episode 300, after which TD+UNC and GATE extend the most consistent leads.

Table 4.2. *CartPole-v1 final and best evaluation reward statistics (10 seeds). Bold indicates best per column. IQR = [Q1, Q3]. MW p = Mann-Whitney U test p-value vs TD baseline*

Condition	Final Mean	Final Std	Best Mean	Median	IQR	MW p	Rank
TD	215.4	86.1	317.3	207.9	[201, 243]	—	6
TD+NOV	275.4	115.0	337.3	269.5	[196, 358]	0.241 ns	3
TD+UNC	320.3	122.9	412.6	348.1	[241, 417]	0.054 ns	1
FULL	264.8	144.2	363.1	263.7	[153, 337]	0.571 ns	4
MULT	248.9	119.5	325.8	255.8	[177, 293]	0.571 ns	5
GATE	276.2	85.7	338.0	273.5	[225, 339]	0.096 ns	2

Mann-Whitney U tests reveal no statistically significant differences between any augmented condition and TD, at either the uncorrected ($p < 0.05$) or Bonferroni-corrected threshold. TD+UNC produces the largest directional effect relative to TD ($p = 0.054$), narrowly missing uncorrected significance. The substantial within-condition variance (standard deviations of 86 to 144 reward units) reflects the sensitivity of bootstrapped DQN to early weight initialisation. Despite the absence of statistical significance, the directional consistency across five of five augmented conditions outperforming TD is notable.

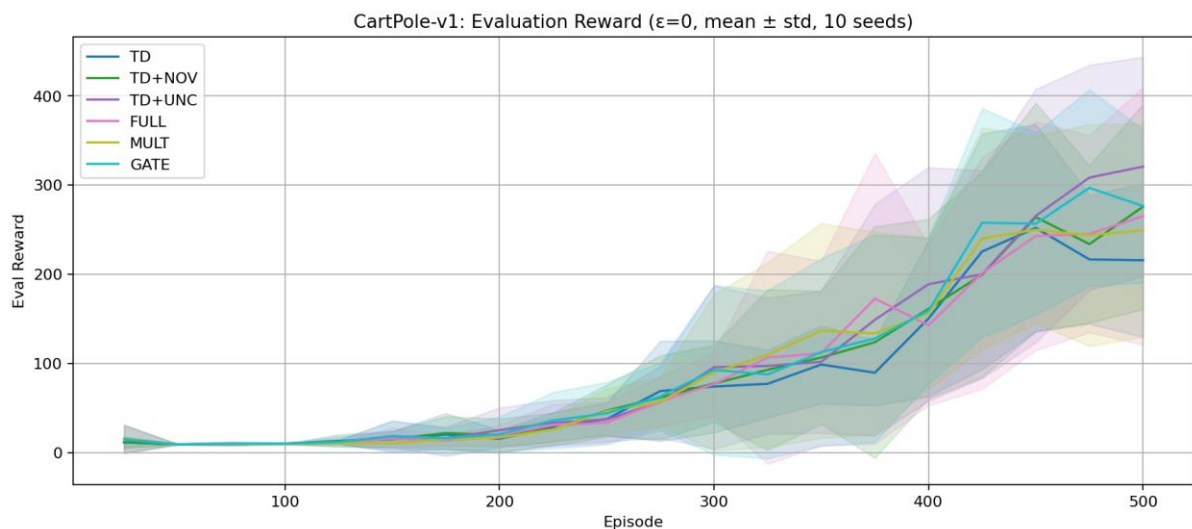


Figure 4.2. *CartPole-v1 evaluation reward ($\epsilon = 0$, mean $\pm$ std, 10 seeds). All conditions show upward trajectories from approximately episode 200 onward. The spread between conditions widens after episode 400.*

4.2.3 Seed Variability

Figure 4.3 shows the per-seed heatmap. Variance within conditions is large. FULL seed 6 reaches the maximum possible score (500.0), as does MULT seed 6 and TD seed 6. Conversely, FULL seed 3 scores only 50.2 and MULT seed 7 only 55.0. This bimodal pattern across seeds — some seeds learning well, others barely improving — is consistent across all conditions and reflects the stochastic nature of early policy formation rather than systematic differences between conditions. GATE has the tightest standard deviation of any augmented condition (± 85.7), and its IQR [224.8, 338.8] is the narrowest, suggesting the gating mechanism provides some regularisation in dense-reward settings.

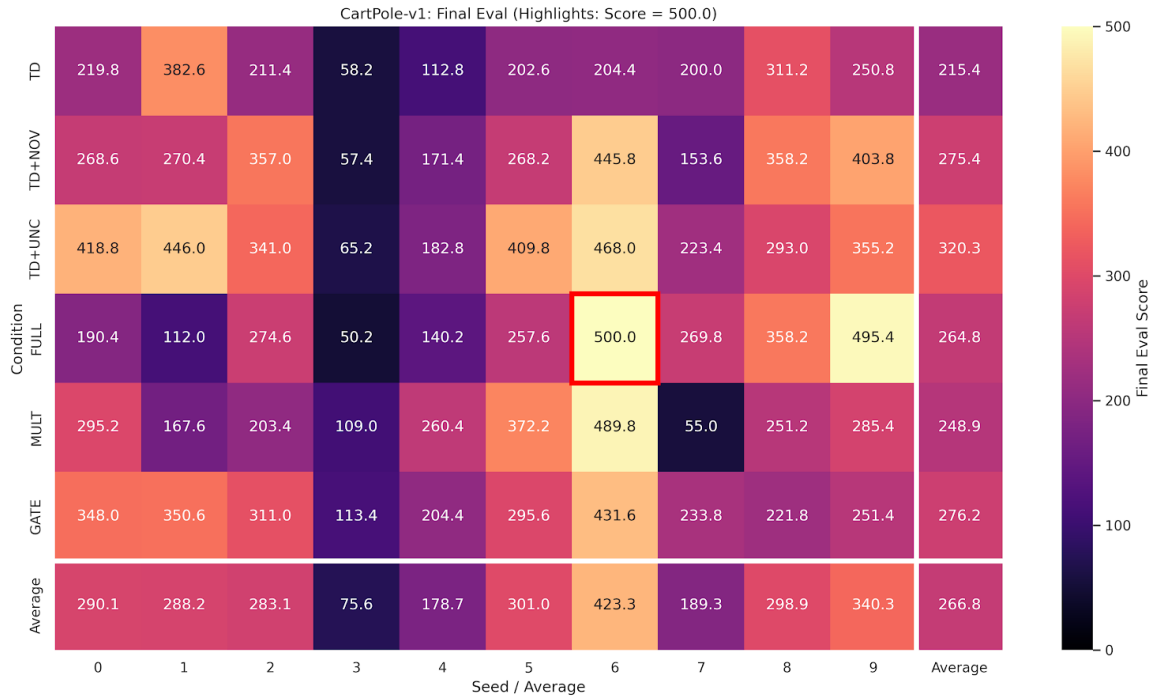


Figure 4.3. CartPole-v1 per-seed final evaluation scores (10 seeds). Each cell represents one seed run; colour encodes score magnitude. Rightmost column shows cross-condition mean per seed (best performing: TD+UNC); bottom row shows cross-seed mean per condition (best: seed 6). The cell highlighted shows the maximum possible score of 500.0.

4.3 MountainCar-v0

4.3.1 Overview

MountainCar-v0 is the primary sparse-reward testbed (see Section 3.7.2). All conditions remain at -200 for the first 325 episodes (mean across seeds), reflecting the environment’s resistance to undirected exploration. Despite exploration (ϵ) being driven by the bio-RPE, it is still insufficient to guide learning within practical time horizons.

4.3.2 Training Reward

Figure 4.4 shows the training reward curves. During training, all conditions remain clustered near the baseline floor of -200 for the first 300-350 episodes. This is due to the binary nature of MountainCar: if the goal is not reached, a minimum of -200 reward units is returned, providing no gradient to distinguish conditions. The first deviations above -200 emerge sporadically from episode 50 onward in GATE and from episode 100 in TD, with most

conditions beginning a slow, noisy ascent after episode 500. By episode 700, training rewards settle approximately in the -174 to -180 range across conditions.

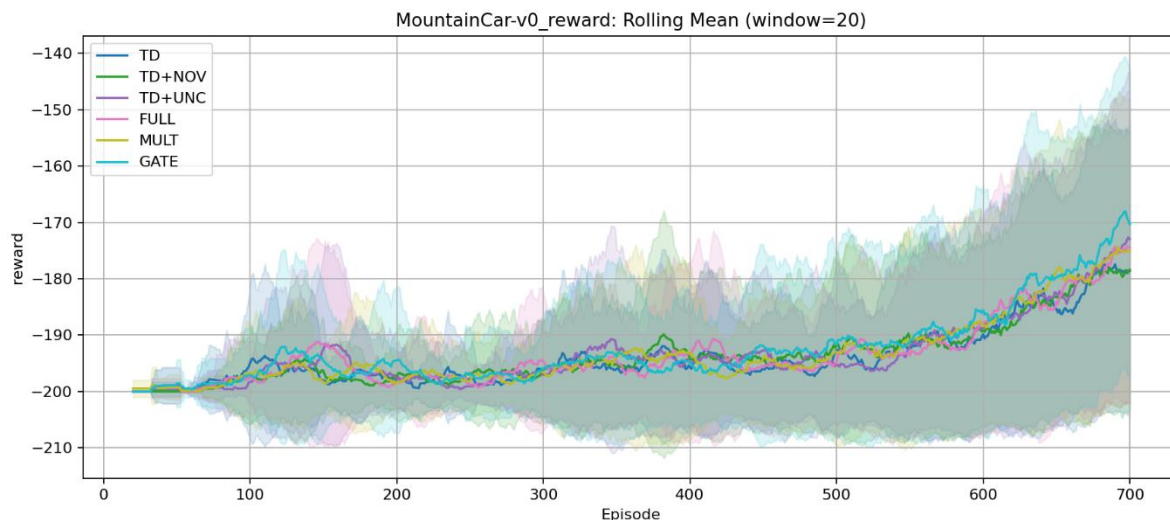


Figure 4.4. MountainCar-v0 training reward (rolling mean, window = 20, mean $\pm$ std, 10 seeds). Curves begin at Episode 20, when the first full rolling average becomes available.

4.3.3 Evaluation Performance

The evaluation reward curve (Figure 4.5) shows similar trends to the training curve. Episodes 1-300 have all conditions at the floor of -200, slowly increasing in episodes 300-500, after which the conditions diverge, where GATE and FULL are shown to be the highest at episode 700.

Table 4.3 presents the MountainCar evaluation statistics. All augmented conditions except TD+UNC (-175.8 ± 24.5 , median -182.2) outperform the TD baseline on final evaluation mean (-169.8). Both GATE (-160.5 ± 20.5) and FULL (-160.6 ± 28.0) tie in the best final mean, with only 0.1 difference in rewards, followed closely by TD+NOV (-162.4 ± 27.7).

GATE has the best peak performance of any condition across all runs (best eval -144.5), has only 1 stuck seed (seed 7) and its IQR $[-165, -148]$ is the tightest of any augmented condition. By contrast, TD+UNC performs the worst (-175.8 ± 24.5 , median -182.2), with 2 stuck seeds and the lowest median.

Table 4.3. MountainCar-v0 final and best evaluation reward statistics (10 seeds). Bold values indicate the best per column. Higher values are better. ns = $p \geq 0.05$. *TD+NOV seed 9 (-199.6) is treated as near-failure.

Condition	Final Mean	Final Std	Best Mean	Median	IQR	Stuck seeds	MW p	Rank
TD	-169.8	18.0	-163.7	-168.2	$[-179, -153]$	1	—	4
TD+NOV	-162.4	27.7	-160.2	-155.9	$[-190, -144]$	2*	0.545 ns	2
TD+UNC	-175.8	24.5	-165.7	-182.2	$[-191, -172]$	2	0.363 ns	5
FULL	-160.6	28.0	-150.3	-154.8	$[-183, -145]$	2	0.427 ns	1
MULT	-165.2	24.7	-149.5	-165.8	$[-188, -148]$	1	0.705 ns	3
GATE	-160.5	20.5	-144.5	-157.4	$[-165, -148]$	1	0.290 ns	1

No pairwise comparison reaches statistical significance on MountainCar (all $p > 0.29$). The inter-condition differences (~15 reward units between best and worst augmented condition) are smaller than the within-condition standard deviations (18-28 reward units), making statistically confident ranking impossible at $n=10$ seeds. The results are therefore interpreted as directional trends: GATE is the highest performing with the most consistent low variance and the strongest peak performance, with FULL and TD+NOV following closely in terms of final mean.

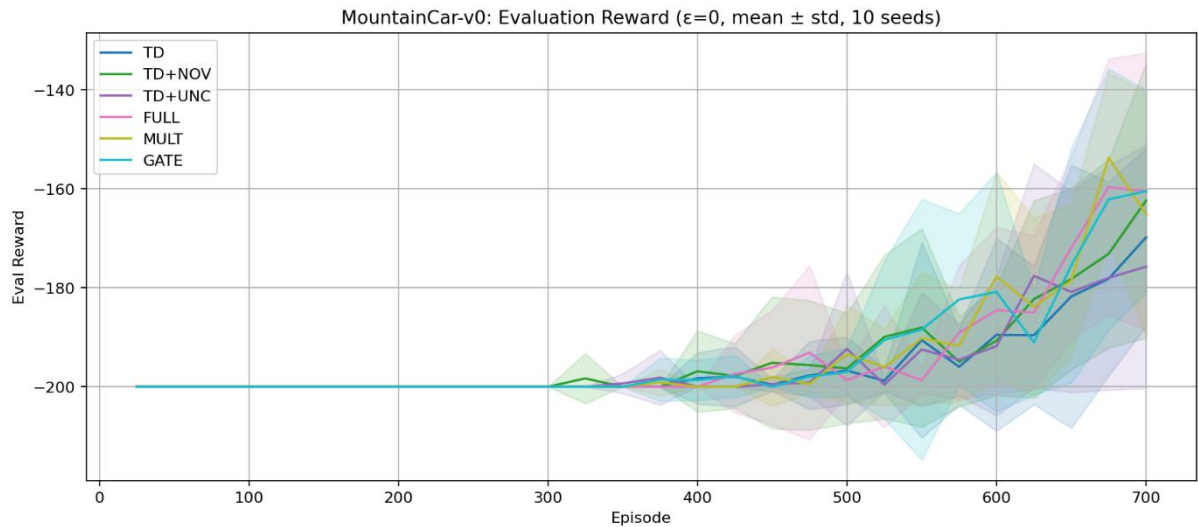


Figure 4.5. MountainCar-v0 evaluation reward ($\epsilon = 0$, mean $\pm$ std, 10 seeds). TD+NOV achieves the best final evaluation reward.

4.3.4 Failure Analysis and State Visitation

Table 4.6 documents all stuck seeds. The additive conditions (TD+NOV, TD+UNC, FULL) each have 2 seeds that failed to discover the goal, while the other conditions (GATE, MULT, TD) only had 1. Additive conditions appear to increase the exploratory variance, as shown in Table 4.3, where the three achieved the biggest standard deviations. This suggests that while additional novelty or uncertainty can help certain runs discover successful trajectories, excessive or poorly targeted exploratory pressure may also delay the consolidation of the precise momentum-based policy required for consistent success. By contrast, multiplicative and gated formulations may reduce this by constraining or dynamically scaling the intrinsic signals' influence.

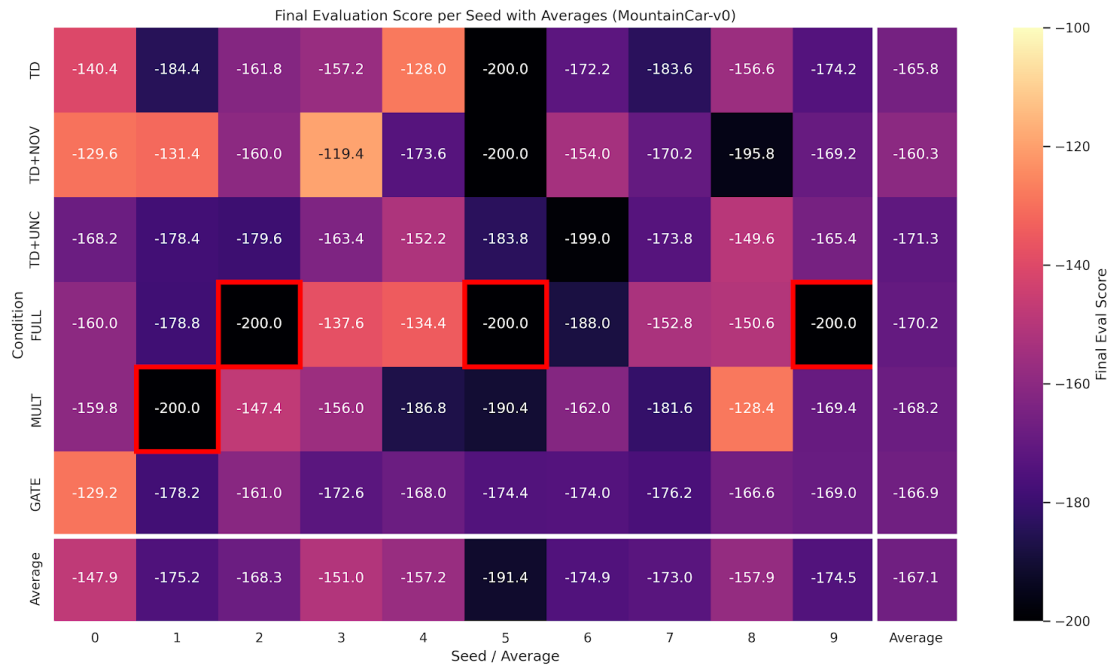


Figure 4.9. MountainCar-v0 per-seed final evaluation scores (10 seeds). Seeds at -200.0 (highlighted with a red border) indicate failure to discover the goal within the 700-episode budget.

4.4 Acrobot-v1

4.4.1 Overview

Acrobot-v1 is a moderately sparse-reward task (see Section 3.7.3). Goal discovery occurs within 100-150 episodes for most seeds, meaning that condition differences reflect convergence quality and stability rather than exploration capability. The six-dimensional state space encodes joint angles and angular velocities, making Acrobot a useful complement to MountainCar for evaluating how well each condition maintains stable training in a richer state space.

4.4.2 Training Reward

The Acrobot training curves (Figure 4.13) are qualitatively different from both CartPole and MountainCar. All conditions start at approximately -470 reward units at episode 1 — near the worst possible performance — and improve rapidly and near-simultaneously between episodes 1 and 150, reaching near -100 by episode 200. After episode 150, all curves plateau and converge in a band between approximately -90 and -130 .

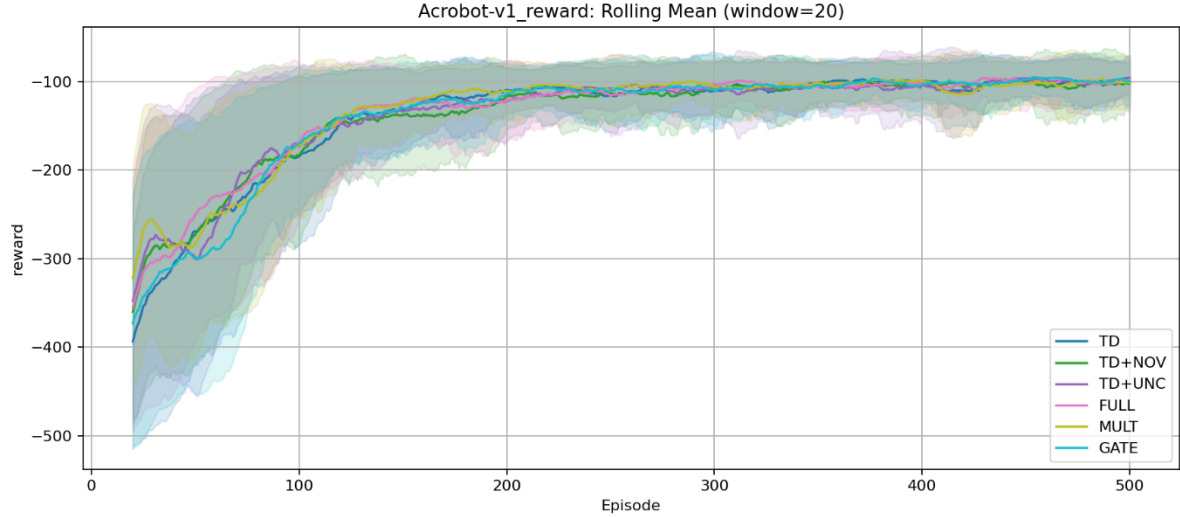


Figure 4.13. Acrobot-v1 training reward (rolling mean, window = 20, mean $\pm$ std, 10 seeds).

4.4.3 Evaluation Performance

Table 4.4 presents the Acrobot results. FULL performs exceptionally well and achieves the best results (-83.9 ± 5.1 , best mean -75.9 , median -82.7 , zero crashes, and IQR $[-88, -81]$) with all ten seeds clustered between -74.8 and -92.8 . This is the tightest seed distribution of any condition across all three environments, and its $p=0.026$ trending significance is the strongest statistical signal in the entire study.

GATE ranks second (-86.7 ± 8.6) and MULT third (-90.4 ± 8.7), both with zero crashed seeds. The TD baseline has the worst final mean (-106.9 ± 28.5) driven by two large outliers (seed 3: -142.2 , seed 6: -166.2).

Table 4.4. Acrobot-v1 final and best evaluation reward statistics (10 seeds). Bold values indicate the best per column. Higher values are better.

Condition	Final Mean	Final Std	Best Mean	Median	IQR	Crashes < -150	MW p	Rank
TD	-106.9	28.5	-79.1	-96.4	$[-126, -87]$	2	—	6
TD+NOV	-99.9	29.0	-79.6	-88.0	$[-94, -85]$	1	0.385 ns	4
TD+UNC	-97.3	29.6	-77.0	-84.8	$[-99, -83]$	1	0.385 ns	5
FULL	-83.9	5.1	-75.9	-82.7	$[-88, -81]$	0	0.026 *	1
MULT	-90.4	8.7	-77.1	-91.1	$[-95, -84]$	0	0.241 ns	3
GATE	-86.7	8.6	-78.3	-85.8	$[-94, -80]$	0	0.130 ns	2

FULL vs. TD is the only comparison in the entire study to reach uncorrected significance ($U=80$, $p=0.026$). Even so, it does not survive Bonferroni correction, and should be interpreted cautiously. The result is consistent with the mean and standard deviation comparison: FULL's tight distribution ($\sigma = 5.1$) versus TD's wide distribution ($\sigma = 28.5$) produces a large rank-sum difference that the Mann-Whitney test detects.

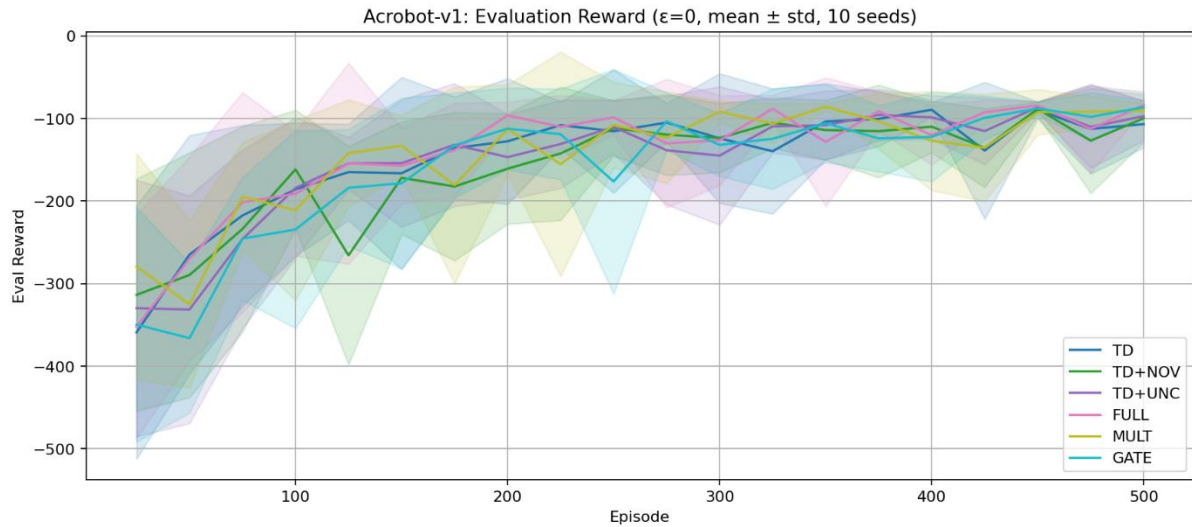


Figure 4.12. Acrobot-v1 evaluation reward ($\epsilon = 0$, mean $\pm$ std, 10 seeds). Most conditions converge within ~ 200 episodes.

4.4.3 Crashed Seeds and Stability Analysis

The absence of crashed seeds under FULL, MULT and GATE suggests that combining novelty and uncertainty improves the robustness across random seeds. The TD baseline is more vulnerable to poor early trajectories because it depends only on external TD error, while TD+NOV and TD+UNC provide only partial correction. By contrast, the combined conditions offer both exploratory breadth and uncertainty-based refinement. This likely explains why FULL, MULT, and GATE produced no crashed seeds, while TD, TD+NOV, and TD+UNC still showed failures.

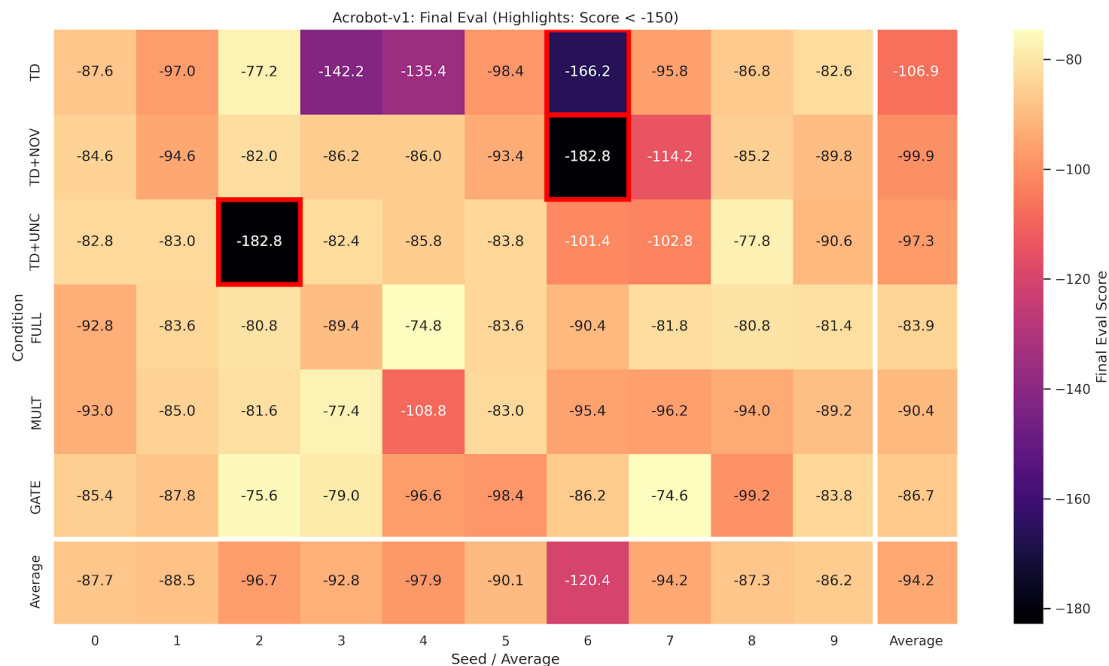


Figure 4.14. Acrobot-v1 per-seed final evaluation scores (10 seeds). Outlier seeds are visually distinct from the tight cluster of successful seeds and highlighted in red.

4.5 Cross-Environment Summary

Table 4.5 consolidates all evaluation statistics and crash counts across environments. Bold values indicate the best result per metric per environment.

Table 4.5. Consolidated evaluation statistics across all environments (10 seeds). CP=CartPole, MC=MountainCar, Ac=Acrobot. Crashes = total seeds stuck at minimum reward. Higher values are better throughout; MC and Acrobot results are inherently negative.

Cond.	CP Final	CP Best	MC Final	MC Best	Ac Final	Ac Best	Crashes	Overall
TD	215.4	317.3	-169.8	-163.7	-106.9	-79.1	3	Baseline
TD+NOV	275.4	337.3	-162.4	-160.2	-99.9	-79.6	2	Best MC, robust
TD+UNC	320.3	412.6	-175.8	-165.7	-97.3	-77.0	3	Best CP, poor MC
FULL	264.8	363.1	-160.6	-150.3	-83.9	-75.9	2	Best Acrobot
MULT	248.9	325.8	-165.2	-149.5	-90.4	-77.1	1	Stable
GATE	276.2	338.0	-160.5	-144.5	-86.7	-78.3	1	Best MC peak

Across environments, no single condition is universally dominant. GATE, however, does demonstrate the most consistent cross-environment competitiveness. It consistently ranks near the top in final evaluations (CartPole: 2nd; MountainCar: 1st, tied; Acrobot: 2nd), while maintaining only one total crash seed, succeeding as the best all-round architecture in terms of adaptability and reliability. For environments with sparser rewards, FULL is the highest performer, ranking 1st in both MountainCar and Acrobot. It demonstrates near-perfect achievements in Acrobot, with best final and best evaluation metrics, tightest seed distribution, zero crashes, and the only statistically significant improvement over TD in the entire study. TD+UNC leads on CartPole but is the worst augmented condition on both sparse-reward environments – a pattern that was consistent across preliminary experimental runs with these hyperparameters.

Table 4.6. All failed and stuck seeds across conditions and environments.

Condition	Environment	Failed Seeds	Notes
TD	MountainCar	Seed 5 (-200.0)	Single stuck seed
TD	Acrobot	Seeds 3 (-142.2), 6 (-166.2)	Inflates baseline mean to -106.9
TD+NOV	MountainCar	Seed 3 (-200.0); Seed 9 (-199.6)	One stuck; one near-failure
TD+NOV	Acrobot	Seed 6 (-182.8)	Single crash
TD+UNC	MountainCar	Seeds 5, 9 (both -200.0)	Two stuck seeds
TD+UNC	Acrobot	Seed 2 (-182.8)	Single crash
FULL	MountainCar	Seeds 2, 5 (both -200.0)	Two stuck seeds
MULT	MountainCar	Seed 3 (-200.0)	Single stuck seed
GATE	MountainCar	Seed 7 (-200.0)	Single stuck seed

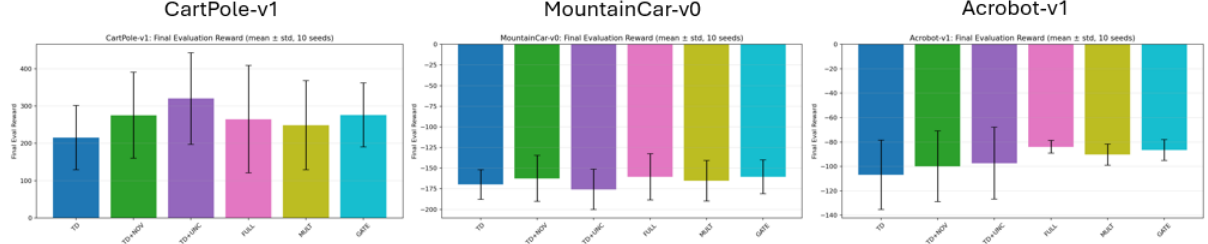


Figure 4.16. Final evaluation reward (mean $\pm$ std, 10 seeds) for all three environments. Error bars show standard deviation across seeds. Conditions are in order: TD, TD+NOV, TD+UNC, FULL, MULT, GATE

4.6 Internal Signal Dynamics

Internal Signals (δ_{bio} , δ_{unc} , δ_{nov} , $\alpha(t)$, $\beta(t)$, ε and lr_{scale}) that characterise this bio-RPE framework were recorded and plotted.

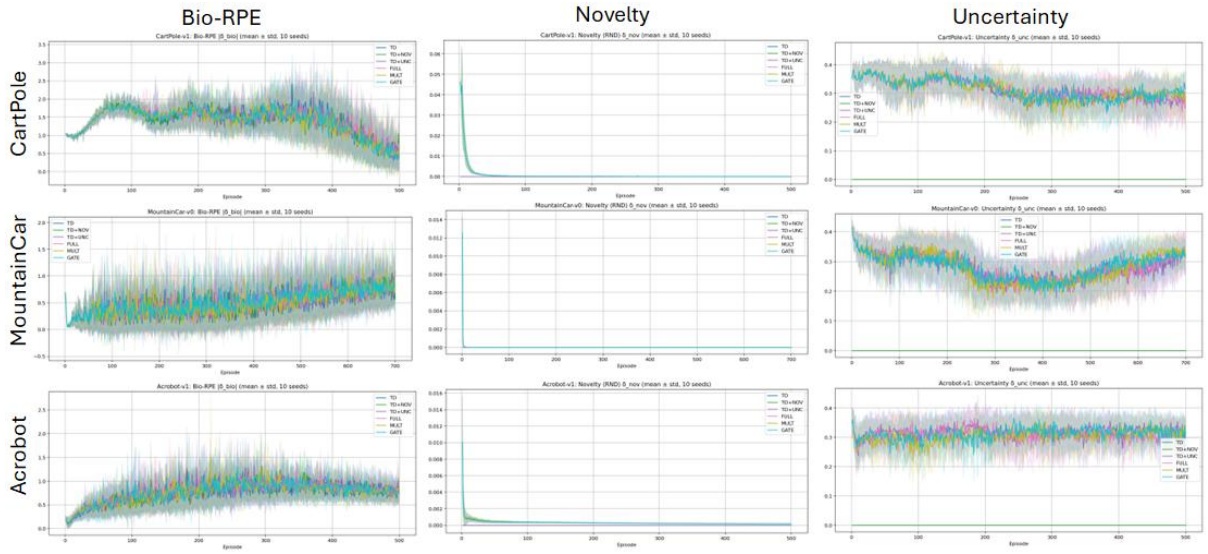


Figure 4.17. Internal signal dynamics across three environments. Left: bio-RPE magnitude. Middle: novelty signal (δ_{nov}). Right: uncertainty signal (δ_{unc}).

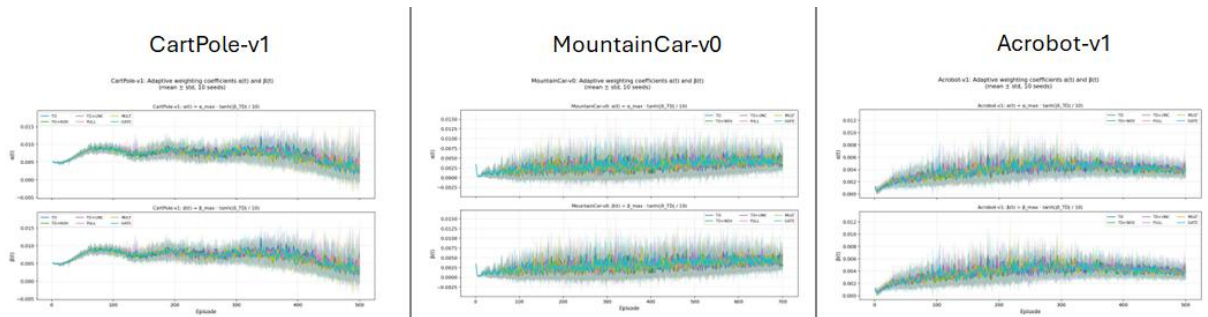


Figure 4.18. Adaptive weighting coefficients $\alpha(t)$ and $\beta(t)$ across training (mean over 10 seeds, per environment). Coefficients scale proportionally with $|\delta_{TD}|$ via the tanh transformation.

4.6.1 Novelty Signal (δ_{nov})

The RND novelty signal exhibits a near-identical decay profile across all three environments. Saturation is especially fast in MountainCar, where δ_{nov} begins at 0.0126, collapses to 0.000085 by episode 5, and reaches functional zero by episode 20. Decay is

slower for Acrobot and CartPole, declining to ~ 0.0005 by episode 50. Acrobot's δ_{nov} persists at 0.0004 through episode 100, while CartPole's decays to 0.0002. This environment-dependent behaviour reflects the differences in state space coverage rate. CartPole's bounded four-dimensional state space is explored more slowly than MountainCar's two-dimensional position-velocity space, leaving the RND predictor with more to learn for longer.

Despite rapid saturation on MountainCar, δ_{nov} provides a meaningful early-training ignition effect: the novelty signal is strongest precisely during the first 5 episodes when the agent's initial trajectory determines which regions of state space are first visited. The plot2_alpha_beta plots (Figure 4.18) confirm that the adaptive weighting coefficients $\alpha(t)$ and $\beta(t)$ respond proportionally to the TD error magnitude, scaling up during high-surprise episodes and approaching zero as predictions stabilise.

4.6.2 Uncertainty Signal (δ_{unc})

The uncertainty signal behaves differently across environments in ways that explain much of the cross-environment performance variation. On MountainCar-v0, δ_{unc} begins at 0.41 and remains persistently elevated throughout the 700-episode window, fluctuating between 0.20 and 0.34 without a clear downward trend. This reflects ongoing ensemble disagreement in the unexplored goal region: because Q-value estimates never fully converge, the uncertainty signal remains an undirected persistent source of bio-RPE escalation. On Acrobot-v1, δ_{unc} begins at 0.35 and stabilises around 0.27–0.33 across training — slightly lower and more stable than MountainCar, consistent with the ensemble reaching partial convergence after goal discovery. On CartPole-v1, δ_{unc} begins high (~ 0.37 at episode 50) and remains broadly stable around 0.27–0.37 throughout, neither growing nor shrinking substantially.

4.6.3 Bio-RPE Magnitude ($|\delta_{bio}|$)

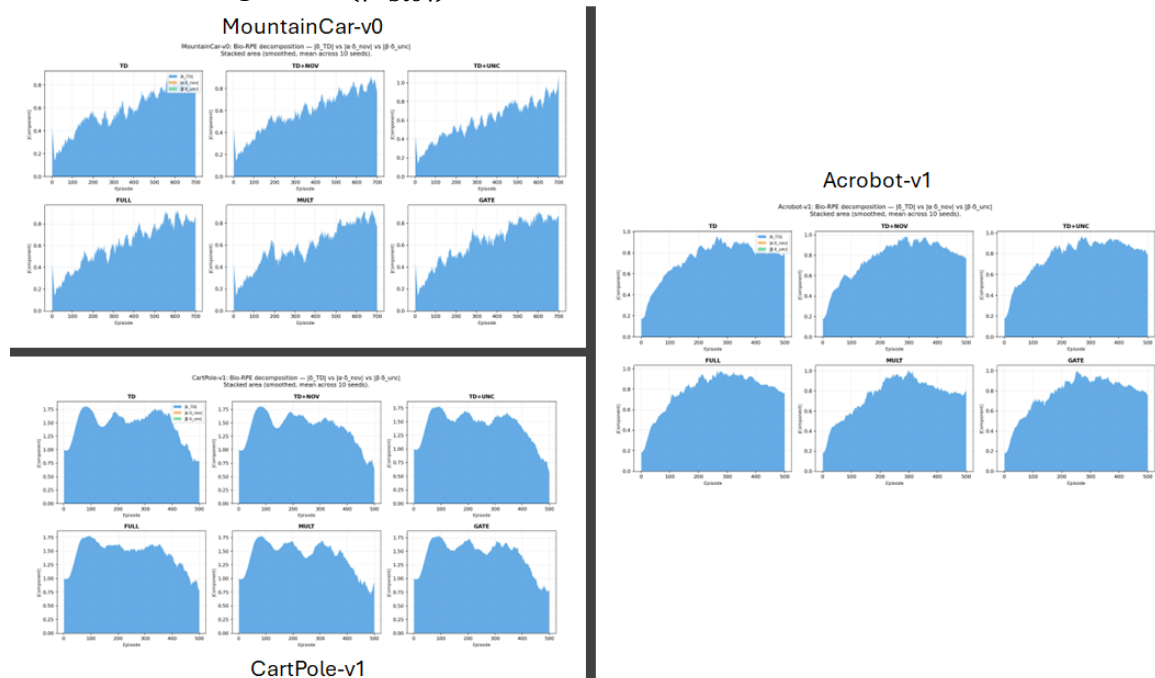


Figure 4.19. Bio-RPE decomposition plots showing the relative contributions of TD error, novelty, and uncertainty to the bio-RPE across training (per environment, mean over applicable conditions).

The bio-RPE trajectories (Figure 4.17, left column) are the most environment-discriminating of all internal signals. On CartPole-v1, bio-RPE begins at approximately 1.48 (episode 50), remains elevated around 1.75–1.90 through episode 300, then declines sharply in the final 100–200 episodes as policies converge: by episode 500, values range from 0.38 (TD+UNC) to 0.65 (MULT), reflecting successful policy stabilisation. The faster decline of bio-augmented conditions relative to TD in the final phase is consistent with their intrinsic signals decaying as the environment becomes familiar, driving ϵ lower and accelerating the transition to exploitation.

On Acrobot-v1, the bio-RPE shows an inverted-U shape: starting at approximately 0.20 at episode 1, rising to peak values of 0.69–1.01 around episode 100–300 as the agent's Q-value estimates undergo rapid revision following goal discovery, then declining to 0.73–0.87 by episode 500. This rise-then-fall profile — not seen in CartPole or MountainCar — reflects Acrobot's distinctive learning dynamics: initial slow exploration, a rapid discovery phase generating large TD errors, and subsequent convergence.

On MountainCar-v0, the bio-RPE shows no convergence pattern: beginning at approximately 0.68 at episode 1, fluctuating around 0.10–0.72 through episode 300, then escalating toward 0.60–1.09 by episode 700. This escalating profile contrasts sharply with the declining CartPole and inverted-U Acrobot trajectories, reflecting the ongoing large TD errors generated by continuous Q-value revision in an environment where the agent has not converged. Importantly, this divergence emerges without any hard-coded environment-specific logic — the bio-RPE responds naturally to each environment's learning dynamics, making it a meaningful emergent diagnostic of convergence status (discussed further in Section 5.8.1).

It is important to note that the bio-RPE and TD error curves are qualitatively similar in most conditions throughout training. This is a direct consequence of the conservative weighting coefficients ($\alpha_{max} = \beta_{max} = 0.05$), which bound the intrinsic contributions to at most 5% each of the TD error magnitude. The bio-RPE decomposition plots (Figure 4.19) confirm that the TD error component dominates throughout training. The significance of the bio-RPE therefore lies not in its numerical divergence from δ_{TD} , but in the directional perturbation it introduces during critical early training phases. This point is discussed further in Section 5.3.

4.6.4 Exploration parameter ϵ

The ϵ curves show stable adaptive exploration across all environments, with exploration highest early in training and gradually shifting toward exploitation as learning progresses. In CartPole, ϵ decays rapidly, reflecting fast convergence in a dense-reward setting. In MountainCar, ϵ remains persistently elevated at a low level after its initial drop, indicating sustained exploratory pressure in response to sparse rewards and unresolved uncertainty. Acrobot exhibits an intermediate pattern, with ϵ stabilising at a moderate plateau consistent with its longer discovery and consolidation phase. Overall, differences between conditions are modest, suggesting that bio-RPE modulation primarily adapts exploration to environment difficulty while maintaining stable behaviour.

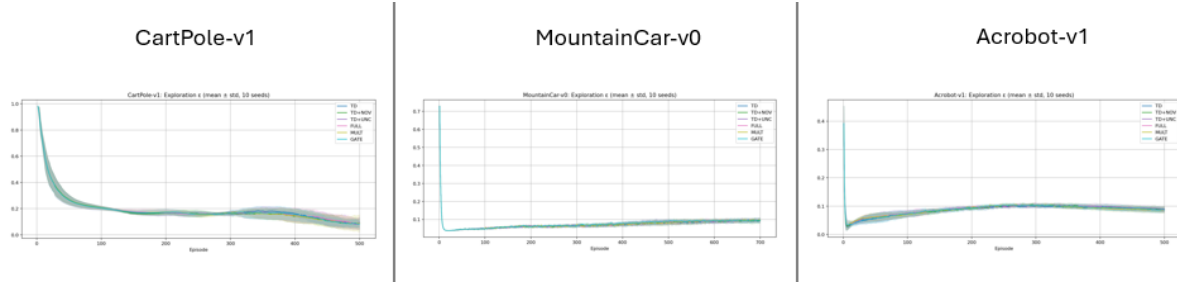


Figure 4.20. Exploration across training (mean over 10 seeds, per environment). Directly derived from $|\delta_{bio}|$;

4.6.5 Learning Rate Scale

The learning rate scale factor (Figure 4.21) follows the bio-RPE profile closely, as it is directly derived from $|\delta_{bio}|$ via the tanh transformation. On CartPole, the LR scale declines in the final 100 episodes as the bio-RPE converges, reducing plasticity as the policy stabilises. On MountainCar, the LR scale shows no decline, maintaining near-maximum values throughout training. On Acrobot, the inverted-U bio-RPE profile produces a corresponding inverted-U LR scale, with the highest plasticity during the goal discovery phase. These patterns confirm that the bio-RPE framework dynamically regulates learning plasticity in a manner that qualitatively mirrors the role of dopamine in modulating synaptic gain (Frémaux & Gerstner, 2016).

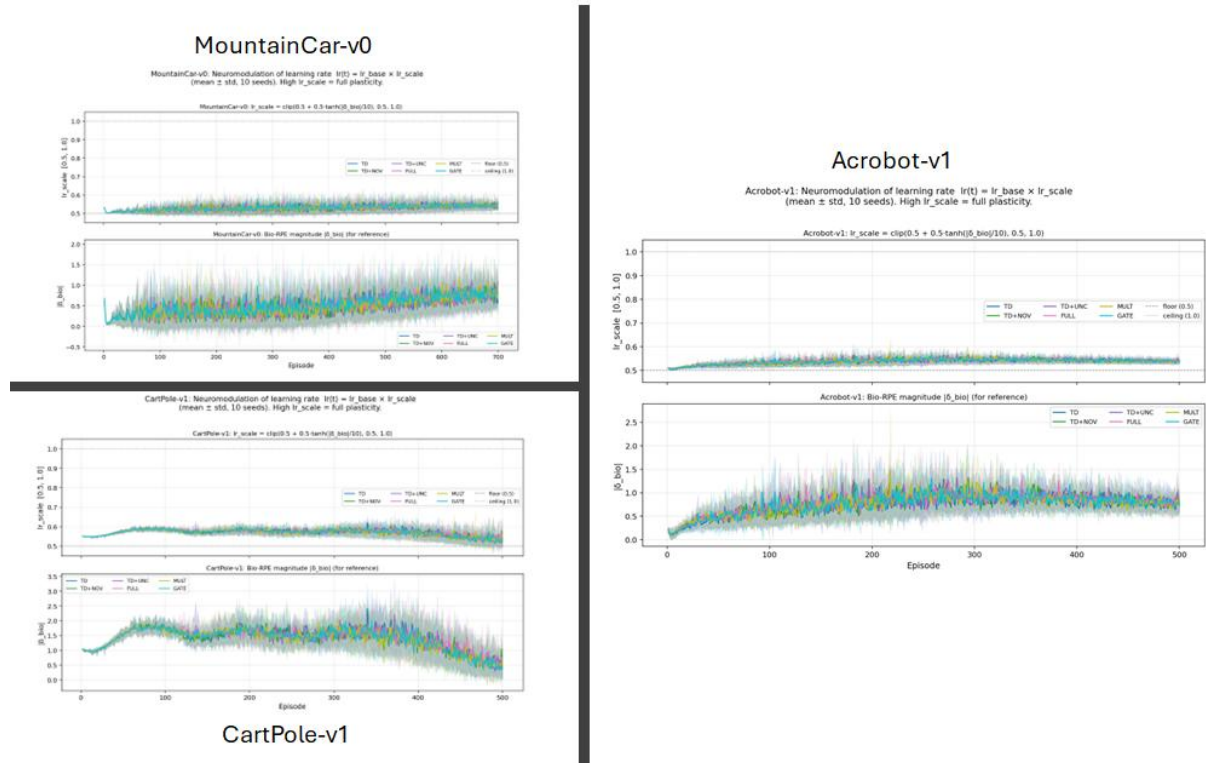


Figure 4.21. Learning rate scale factor across training (mean over 10 seeds, per environment). Directly derived from $|\delta_{bio}|$;

4.6.6 Novelty vs. State Visitation

The novelty-versus-visitation plots (Figure 4.22) illustrate the relationship between RND prediction error and cumulative state visit counts across training. In all environments, the novelty signal is highest in states that have been visited least frequently, confirming that the RND architecture is correctly identifying unvisited regions.

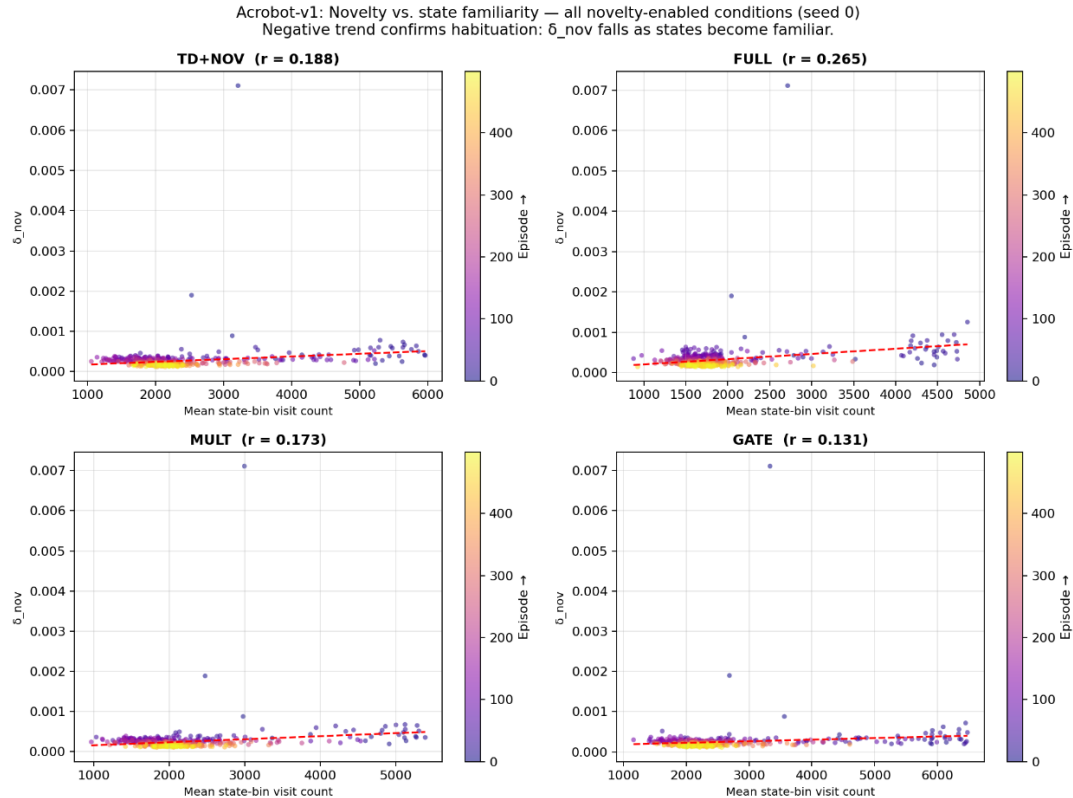


Figure 4.22. Novelty signal magnitude as a function of state visitation frequency. Novel (infrequently visited) states generate higher δ_{nov} , confirming the RND architecture is operating correctly.

Figure 4.23 compares the state visitation heatmaps between TD and TD+NOV. The comparison suggests that novelty modestly broadens state visitation while preserving strong concentration around the central stabilisation region (cart position ≈ 0 , pole angle ≈ 0). The agent still spends most of its time near equilibrium, but TD+NOV shows slightly wider coverage of peripheral and transitional states, indicating that novelty encourages broader state sampling rather than narrowly repeating familiar trajectories. In CartPole's dense-reward setting, novelty does not fundamentally alter the learned control structure, but instead appears to function primarily as an exploration-diversification mechanism that expands state exposure while maintaining policy stability.

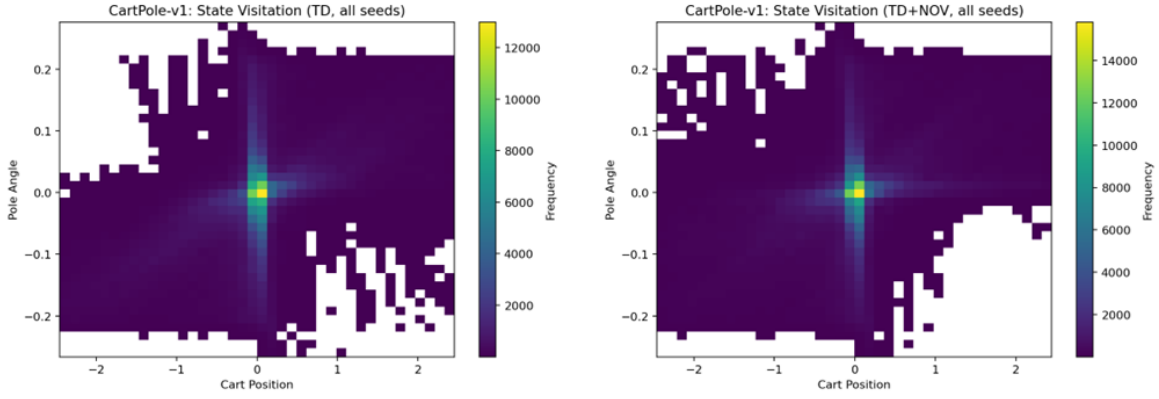


Figure 4.23. State Visitation heatmaps for TD baseline (left) and TD+NOV (right).

5. Discussion

This section interprets the findings in relation to the five research questions.

5.1 RQ1: Mathematical Integration of Reward, Novelty, and Uncertainty

The three integration formulations – additive (linear synaptic summation), multiplicative (gain modulation), and gated (epistemic transition) – are described in full in Section 3.2, where their biological motivations are established. RQ1 asks whether these formulations are mathematically coherent and produce stable training dynamics in practice.

Empirically, all six conditions trained without numerical instability across all 180 runs. The bio-RPE signal remains well-formed throughout, and the adaptive weighting coefficients $\alpha(t)$ and $\beta(t)$ respond appropriately to the magnitude of the TD error as they scale up during high-surprise episodes and decay as predictions stabilise (Figure 4.18). This behaviour mirrors the dopamine gain modulation described by Schultz (2016), where larger prediction errors elicit stronger modulation of novelty and uncertainty signals.

The conservative bound on α and β ($\alpha_{max} = \beta_{max} = 0.05$) was a critical design choice. Because intrinsic contributions are capped at 5% of the TD error magnitude, the bio-RPE decomposition plots (Figure 4.19) confirm that δ_{TD} dominates throughout training. This means the bio-RPE is not a wholesale replacement of the temporal-difference signal but a directional perturbation of it. The system remains anchored to extrinsic reward while being augmented by biologically grounded signals during critical early phases. RQ1 is therefore answered affirmatively: all three integration modes are mathematically coherent, produce well-behaved training dynamics, and satisfy the biological motivations from which they were derived.

It is important to note that the behaviour of the additive formulation depends on the normalisation of its components. Novelty provides a brief “ignition” signal before saturating, leaving the bio-RPE signal to be driven by uncertainty in later training. This imbalance between short-term novelty and persistent uncertainty has significant implications for cross-environment performance (see RQ4).

5.2 RQ2: Optimal Integration Scheme for Exploration and Stability

RQ2 asks which integration scheme best balances exploration and stability. Across the three environments, no single scheme achieves universal superiority. However, the gated and additive (FULL) formulations emerge as the most consistently competitive.

5.2.1 FULL – Best on Acrobot, tied 1st on MountainCar

The additive formulation applies novelty and uncertainty as unconditional additives to the bio-RPE. On Acrobot, this works well because the two signals naturally complement each other across training: novelty is strongest in early episodes when the state space is unfamiliar, then fades as novelty saturates, at which point uncertainty takes over during the rapid Q-value revision phase. This natural handoff explains FULL's exceptionally tight seed distribution ($\sigma=5.1$). The joint signal essentially prevents early-trajectory trapping that caused TD's two crashes. On MountainCar, the weakness of unconditional addition becomes apparent: once novelty saturates by episode 20, uncertainty becomes the sole contributor but remains persistently high and undirected throughout training, providing pressure without pointing the agent anywhere useful.

5.2.2 GATE – Best all-rounder, best peak on MountainCar

The sigmoid gate is GATE's key architectural advantage. The sigmoid gate (Section 3.2.3) dynamically re-weights which signal is active based on the agent's current epistemic state, meaning it never wastes bio-RPE on a saturated novelty signal when uncertainty is more informative, and vice versa. This explains both GATE's best peak on MountainCar (−144.5) and its tighter IQR [−165, −148] versus FULL's [−183, −145]: the gate suppresses exploratory pressure in states the agent already understands, reducing overshooting. The main limitation is that the gate threshold τ and steepness k are fixed hyperparameters rather than learned, so the transition point is not adapted to the environment — which likely explains why GATE doesn't consistently beat FULL on Acrobot, where FULL's simpler summation happens to align well with the training dynamics.

5.2.3 MULT (Multiplicative, both signals) — Consistent mid-tier

The multiplicative scheme scales δ_{TD} by $(1 + \alpha \cdot \tilde{n} \cdot nov + \beta \cdot \tilde{u} \cdot unc)$ rather than adding to it. With $\alpha_{max} = \beta_{max} = 0.05$, this means the bio-RPE can be at most 10% above the raw TD error — a very narrow range. Unlike FULL and GATE, MULT can never produce a bio-RPE signal that substantially exceeds δ_{TD} , so it rarely drives ε or LR meaningfully differently from the baseline. This explains its placement: stable enough to avoid crashes (only one stuck seed total), but without enough headroom to reach the peaks that FULL and GATE achieve when intrinsic signals are strong.

5.2.4 TD+UNC (Uncertainty only) — Best on CartPole, worst augmented on MountainCar

This is the most architecturally revealing pattern in the study. On CartPole, ensemble disagreement is a reliable proxy for genuinely unexplored high-value regions, so the uncertainty signal provides targeted guidance. On MountainCar, δ_{unc} remains elevated (0.20–0.34) across all 700 episodes because the agent never fully converges on the goal region, resulting in the signal becoming persistent background noise rather than useful guidance. Without δ_{nov} to provide early-phase diversity, poorly initialised seeds have no mechanism to break out of suboptimal regions, which are frequent in sparse-reward environments. This resulted in two stuck seeds and performance below the TD baseline.

5.2.5 TD+NOV (Novelty only) — Solid on MountainCar, average elsewhere

RND novelty provides a brief but critical early-training ignition effect on MountainCar: in the first 5–20 episodes, the novelty bonus encourages the agent to explore

regions it would otherwise avoid under a flat reward landscape, giving some seeds the push needed to build momentum toward the goal. However, once δ_{nov} saturates the architecture has no remaining intrinsic signal and behaviour reduces to pure TD — which is why TD+NOV still produces two stuck seeds that the ignition effect couldn't rescue. On CartPole and Acrobot, the rapid saturation matters less because dense rewards or early goal discovery provide the residual learning signal.

5.2.6 TD (Baseline) — Worst on sparse environments

With no intrinsic signal, TD relies entirely on extrinsic reward gradients. On CartPole's dense-reward setting this is sufficient, but on MountainCar and Acrobot the agent depends almost entirely on lucky initialisation to discover the goal, resulting in the most crashed seeds overall and the widest performance distributions.

The clearest architectural takeaway across all conditions is that the combination of novelty and uncertainty matters more than the integration mode. Formulations with both signals - FULL, MULT, and GATE - all produce zero crashes on Acrobot while the partial-signal and baseline conditions do not. The mode determines the ceiling (GATE and FULL reach higher peaks) and the variance (GATE has tighter IQR), but the presence of both signals is what provides the floor of robustness.

However, regarding the specific question of which scheme best balances exploration and stability: GATE is the most reliable all-round choice (CartPole 2nd, MountainCar 1st, Acrobot 2nd; one total crash seed). FULL is preferable in sparse settings where Acrobot-like dynamics dominate and peak performance matters more than risk. MULT is a consistent mid-tier performer but does not clearly outperform either GATE or FULL in any environment. These trade-offs suggest that the optimal scheme is environment-dependent, a finding that is both a result and a limitation of the current study.

5.3 RQ3: Sample Efficiency and Convergence Stability

5.3.1 Sample Efficiency

The evidence for improved sample efficiency is directional but not statistically confirmed at $n = 10$ seeds. On MountainCar, FULL, GATE, and TD+NOV all outperform the TD baseline on final evaluation mean; GATE and MULT lead on best evaluation mean. δ_{nov} is spatially concentrated in the goal-proximal right-side region during the first 5 episodes on MountainCar, biasing early exploration toward those states before collapsing. Once the agent visits goal-proximal states, the TD error updates propagate reward information through the replay buffer and standard Q-learning takes over. The novelty signal functions as a brief directional ignition that seeds the TD error with informative early experiences, without needing to persist throughout training. This interpretation is supported by the bio-RPE decomposition plots (Figure 4.19): the novelty contribution is non-trivially visible only in the earliest episodes, after which the TD error dominates.

The adaptive ϵ modulation also contributes to sample efficiency, though subtly. The U-shaped ϵ profile on MountainCar (Section 4.6.4) means that exploration is sustained as the agent encounters new regions of state space — a property that fixed decay schedules cannot provide. The final episode differentiation on CartPole (Figure 4.21), where TD+UNC and TD+NOV maintain lower ϵ than TD, reflects that faster bio-RPE decline in converging conditions accelerates the transition to exploitation. These dynamics are consistent with the theoretical motivation: the bio-RPE modulates ϵ in response to the agent's actual learning state rather than an externally imposed schedule.

5.3.2 Convergence Stability

The strongest evidence for improved convergence stability comes from FULL on Acrobot. A standard deviation of 5.1 across ten seeds, with all seeds clustered between -74.8 and -92.8 , is substantially tighter than the TD baseline ($\sigma = 28.5$). The Mann-Whitney test result ($p = 0.026$), while not surviving Bonferroni correction, is the strongest statistical signal in the entire dataset. The mechanism is plausible: on Acrobot, where goal discovery is rapid, the combined novelty and uncertainty signals provide a richer early-training gradient that consistently guides all seeds toward the goal region, after which the TD error drives uniform convergence. The zero crashed seeds for FULL, MULT, and GATE on Acrobot further support the stability interpretation. In answer to RQ3: the framework shows directional improvements in both sample efficiency (FULL and GATE leading on MountainCar, with novelty signals as the likely mechanism) and convergence stability (FULL on Acrobot), neither of which is statistically confirmed at $n = 10$ seeds.

5.4 RQ4: Framework Performance Across Sparse and Dense Reward Environments

RQ4 asks how the bio-RPE framework performs across different environment types. First, the framework does not impede dense-reward learning. On CartPole, all augmented conditions outperform TD on final evaluation mean, and several seeds across FULL and MULT reach the maximum possible score (500). The bio-RPE framework is compatible with dense-reward learning because the adaptive weighting mechanism ($\alpha(t)$ and $\beta(t)$ scaling with $|\delta_{TD}|$) naturally reduces intrinsic contributions as the TD error declines with learning, allowing the bio-RPE to approximate standard TD learning in later training for exploitation. However, statistical significance is not reported, and is attributable to the high within-condition variance from initialisation sensitivity rather than an algorithmic flaw. Notably, GATE achieves the tightest standard deviation (± 85.7), consistent with the hypothesis that dynamic gating moderates exploratory variance.

Second, novelty-based exploration provides directional benefits in hard sparse-reward settings. On MountainCar, all conditions with δ_{nov} (TD+NOV, FULL, GATE, MULT) all outperform the TD baseline numerically, while the condition without novelty (TD+UNC) as the worst augmented condition. Novelty-enabled conditions achieve a broader coverage of the position-velocity space (Fig 4.23), whilst conditions without novelty remain trapped in the left valley as exploration with no direction does not help in breaking the momentum barrier. However, no condition achieves statistical significance, and the novelty signal saturates too quickly (within 20 episodes; Section 4.6.1), inheriting the RND’s limitation of saturation. The uncertainty signal remains persistently elevated throughout ($\delta_{unc} \in [0.20, 0.34]$ across all 700 episodes), which produces ongoing bio-RPE perturbation but without the directional specificity required to guide the agent toward the goal region.

In the moderately sparse Acrobot environment, FULL demonstrated superior convergence and stability, achieving the best final evaluation (-83.9 ± 5.1) and the study’s only statistically significant lead over TD ($p=0.026$). Since all agents discovered the goal early, these gains reflect convergence quality rather than raw exploration. Notably, the tenfold reduction in standard deviation (5.1 vs. 28.5) confirms that the combined novelty and uncertainty signal successfully stabilizes the learning process once the goal has been found.

An important and unexpected finding regarding RQ4 is the inversion of TD+UNC’s performance across reward types. TD+UNC achieves the best final and best evaluation scores

on CartPole (320.3 and 412.6, respectively) but is the worst augmented condition on MountainCar (−175.8, worse than TD at −169.8) and ranks fifth on Acrobot (−97.3). This suggests that uncertainty-only augmentation is well-suited to dense reward settings where Q-value disagreement is rapidly resolved and provides accurate exploration guidance but becomes counterproductive in sparse settings where ensemble disagreement persists as an undirected background signal rather than identifying genuinely exploitable uncertainty.

5.5 RQ5: Does the Agent’s Learning Behaviour Resemble Dopaminergic Mechanisms?

RQ5 asks whether the bio-RPE framework's empirical behaviour resembles biological dopaminergic mechanisms. This question operates at two levels: whether the mathematical formulation captures biologically grounded principles, and whether the observed learning dynamics exhibit qualitative patterns consistent with dopaminergic neuroscience.

At the formulation level, the bio-RPE framework explicitly models four neuroscientific mechanisms. First, the tanh-bounded adaptive weighting of $\alpha(t)$ and $\beta(t)$ mirrors phasic dopamine gain modulation, in which stronger prediction errors transiently amplify the salience of concurrent inputs from novelty (hippocampus-VTA pathway) and uncertainty (prefrontal cortex) (Bromberg-Martin et al., 2010). Second, the multiplicative integration mode reflects the Bayesian brain's precision-weighted prediction errors, where the brain scales each error by the estimated precision of the signal, and also the gain-control theory of neuromodulators in basal ganglia circuitry (Friston, 2009; Doya, 2002). Third, the gated integration mode approximates the prefrontal-striatal switching mechanism described by Frank et al. (2007), in which top-down cognitive control allocates processing resources between exploration and uncertainty-resolution pathways depending on the agent's current epistemic state. Fourth, the learning rate modulation directly implements the computational analogue of dopamine-driven synaptic plasticity: elevated bio-RPE increases the Adam learning rate toward LR_{base} , facilitating stronger updates from recent experience, while reduced bio-RPE decreases it toward $\frac{LR_{base}}{2}$, protecting established weights (Frémaux & Gerstner, 2016).

At the empirical dynamics level, several qualitative correspondences are observed. The rapid saturation of the novelty signal ($\delta_{nov} \rightarrow 0$ within 5–20 episodes) parallels the rapid habituation of dopaminergic novelty responses in animals exposed repeatedly to the same stimuli: phasic dopamine bursts to novel stimuli attenuate quickly as the stimulus becomes familiar (Kakade & Dayan, 2002; Wang et al., 2024). The persistent elevation of δ_{unc} in MountainCar's goal region, an area the agent has not learned to reach consistently, mirrors the tonic uncertainty-driven dopamine activity observed when animals face unresolved prediction error, maintaining exploratory pressure without clear directional guidance.

The bio-RPE decay profiles (Figure 4.17) mirror the temporal structure of dopaminergic activity during conditioning. On CartPole and Acrobot, the bio-RPE is elevated during early and mid-training when the agent is actively learning, then declines as predictions stabilise — paralleling the reduction in dopamine responses to reward-predicting stimuli as predictions become accurate (Schultz, 2016; Fiorillo et al., 2003). On MountainCar, the absence of this decline reflects a genuine failure to converge: the bio-RPE remains elevated because the agent's predictions remain inaccurate, an appropriate response that mirrors the sustained dopaminergic activity observed when reward contingencies remain uncertain (Schultz, 2016).

The adaptive weighting coefficients (Figure 4.18) also exhibit the proportional scaling with TD error magnitude that characterises phasic dopamine: $\alpha(t)$ and $\beta(t)$ are large during early training when prediction errors are high and the agent is in a high-surprise regime, and approach zero as Q-value estimates stabilise and the TD error decays. This matches the temporal profile of dopamine modulation across acquisition versus consolidation phases of learning in animal models (Mikhael et al., 2025).

However, important divergences from biological dopaminergic systems also exist. Biological dopamine encodes signed prediction errors (positive bursts for better-than-expected outcomes; pauses for worse-than-expected), whereas the current bio-RPE uses the absolute magnitude of δ_{TD} for modulation. The sign information in δ_{TD} is preserved in the Q-network update but is not used to differentially modulate the exploration or plasticity signals. Furthermore, the current framework does not model the D1/D2 receptor dynamics that mediate opposing effects of dopamine on direct and indirect basal ganglia pathways (Mikhael et al., 2025), nor does it capture the temporal credit assignment across extended behavioural sequences that is central to dopaminergic function. These divergences are acknowledged as limitations (Section 5.6) and represent meaningful directions for future biologically grounded modelling.

5.6 Limitations

Several limitations constrain the conclusions that can be drawn. First, $n = 10$ seeds is insufficient for statistically robust pairwise comparison. The use of 10 seeds per condition reflected a practical computational compromise, as the full 180-run experiment already required over 24 hours on a resource-limited Google Cloud free-trial VM, making larger seed counts infeasible within available time and infrastructure. Due to this, most inter-condition differences are within one standard deviation, and only one comparison (FULL vs TD on Acrobot) reaches even uncorrected significance. A minimum of 20 seeds and stratified bootstrap confidence intervals (Agarwal et al., 2021) would be required for reliable statistical claims. The directional findings reported here should be treated as hypotheses for replication rather than confirmed effects.

Second, no environment-specific hyperparameter tuning was performed. A single set of hyperparameters ($\alpha_{max} = \beta_{max} = 0.05$, gate threshold $\tau = 0.5$, gate steepness $k = 5.0$, RND learning rate 3×10^{-4}) was applied across all conditions and environments. Environment-specific tuning, or adaptive methods such as population-based training (Jaderberg et al., 2017), may substantially improve conditions that were disadvantaged by the fixed setting — particularly MULT and GATE, whose performance is more sensitive to early-training signal magnitudes.

Third, the three environments cover a limited range of exploration difficulty. Hard exploration benchmarks such as Montezuma's Revenge, Pitfall, and MiniGrid are substantially more demanding (Bellemare et al., 2016; Ecoffet et al., 2021) and were not evaluated. Whether the directional advantages observed on MountainCar scale to environments where the goal region is even smaller relative to the state space remains an open question.

Fourth, no comparison against published external baselines was conducted. All comparisons are internal, between the six ablation conditions. Evaluating the bio-RPE framework against standard RND-DQN (Burda et al., 2019), ICM (Pathak et al., 2017), or NGU

(Badia et al., 2020) is necessary to position the contribution within the broader intrinsic motivation literature.

Fifth, the bio-RPE and TD error are numerically close throughout training; a consequence of the conservative α/β weighting. The decomposition plots (Figure 4.19) make this explicit. Realising the full theoretical potential of the unified bio-RPE signal — where novelty and uncertainty contribute substantively rather than marginally — requires either larger weighting coefficients, which preliminary experiments showed can destabilise training, or more sophisticated adaptive weighting schemes.

5.7 Future Work

5.7.1 Framework Scaling, Novelty Extension, and Validation

A key next step is to strengthen the bio-RPE framework through broader environmental testing, improved novelty estimation, and more robust statistical validation. Extending evaluation beyond low-dimensional Gymnasium tasks to higher-dimensional environments such as Atari, MuJoCo, or partially observable domains would test whether the framework’s benefits generalise when exploration challenges are more complex and novelty saturates more slowly. The rapid collapse of RND novelty in MountainCar also suggests that more persistent novelty estimators, such as NGU, episodic curiosity, or count-based methods, may improve sustained exploration in sparse environments. Finally, although ten seeds represented a practical computational compromise, larger seed counts remain necessary for stronger statistical confidence. Expanding to 30+ seeds, or using alternatives such as bootstrap confidence intervals or stratified seed selection, would provide more reliable validation of the directional trends observed here. Together, these extensions would improve the framework’s generalisability, exploration capacity, and methodological rigor.

5.7.2 Bio-RPE as a Training Diagnostics Signal

One of the most promising practical extensions of this framework is the use of bio-RPE magnitude as an unsupervised training diagnostic. Because $|\delta_{bio}|$ naturally produced declining, inverted-U, or escalating profiles depending on convergence status across environments, it may serve as an online indicator of whether learning is stabilising, stagnating, or failing without requiring external evaluation checkpoints. This creates potential applications in adaptive curriculum learning, early stopping, or automated training-budget allocation. For example, environments could automatically increase in difficulty once $|\delta_{bio}|$ falls below a stability threshold, while persistently escalating $|\delta_{bio}|$ could trigger extended training or targeted exploration interventions. The novelty signal’s rapid saturation also suggests an alternative design direction in which novelty functions less as a continuous intrinsic reward and more as a targeted ignition mechanism during early training or periods of detected stagnation.

5.7.3 Signed RPE and D1/D2 Receptor Dynamics

A major biological simplification of the current framework is its use of $|\delta_{TD}|$ magnitude for modulation while discarding sign-sensitive dopaminergic asymmetry. Future work could incorporate signed TD error directly into the modulation functions, allowing positive δ_{TD} to enhance plasticity and exploratory confidence while negative δ_{TD} suppresses maladaptive learning or promotes caution, analogous to D1/D2 receptor pathway dynamics. This would improve biological fidelity and may produce qualitatively different behaviour in asymmetric or punishment-heavy reward environments. Computationally tractable

frameworks such as Mikhael et al. (2025) provide a plausible basis for implementing this extension.

6. Project Management and Reflection

The project followed a structured, phased approach as outlined in the initial Gantt Chart (Figure 6), managed through biweekly supervisor meetings to ensure steady progress and alignment with academic goals.

The first phase (September to December) focused on the core bio-RPE framework implementation within a Bootstrapped DQN, using CartPole for validation and MountainCar for initial testing of TD, novelty, and uncertainty conditions. The second phase (January to April) expanded the study to include MULT and GATE integration schemes, hyperparameter tuning, and the Acrobot environment. A critical pivot occurred when local training on Google Colab proved inefficient; transitioning to a Google Cloud VM was essential to complete the 180 total runs required for the study, as each three-seed batch required approximately seven hours of computation.

This project allowed me to explore a long-term interest in computational neuroscience that began in my first year. A key lesson in project feasibility was the decision to omit Montezuma's Revenge, which ensured a more thorough analysis of existing environments rather than overextending resources. While the statistical insignificance of some results at $n = 10$ seeds was initially discouraging, it provided a realistic perspective on the "Rule of 30" and the inherent noise in sparse-reward environments. Ultimately, the project successfully demonstrated that biologically grounded signals like novelty and uncertainty can improve exploration robustness and convergence stability, providing a functional bridge between neuroscientific principles and reinforcement learning.



Figure 6. Gantt Chart from Project Proposal

7. Conclusion

This project introduced and evaluated a dopamine-inspired reinforcement learning framework that integrates temporal-difference error, novelty, and uncertainty into a unified biological reward prediction error (bio-RPE) signal for adaptive neuromodulation of learning rate and exploration. Across 180 experimental runs spanning dense-reward (CartPole) and sparse-reward (MountainCar, Acrobot) environments, the framework demonstrated that biologically motivated signal fusion is computationally tractable, mathematically stable, and capable of producing meaningful directional improvements over a Bootstrapped DQN

baseline. The three integration schemes revealed distinct trade-offs: FULL produced exceptional convergence stability on Acrobot, GATE provided the strongest cross-environment robustness and best MountainCar peak performance, while novelty-based conditions demonstrated that even rapidly saturating intrinsic signals can meaningfully improve early sparse exploration. More broadly, the results suggest that combining novelty and uncertainty is more important than any single integration mode alone, as dual-signal conditions consistently improved robustness against failed seeds in sparse settings.

Beyond performance, the project's most original contribution may be conceptual: the bio-RPE signal itself exhibited emergent diagnostic properties, naturally reflecting convergence, instability, or unresolved exploration difficulty without external supervision. This positions bio-RPE not only as a learning signal but also as a potential training diagnostic for future adaptive RL systems. While statistical power, limited benchmark complexity, and rapid novelty saturation constrain definitive conclusions, the framework establishes a credible proof of concept for biologically grounded RL architectures that move beyond treating novelty and uncertainty as isolated reward bonuses. In doing so, this work contributes both an engineering framework for adaptive exploration and a computational perspective on how dopaminergic principles may inform more flexible, interpretable reinforcement learning systems. Future extensions to larger-scale environments, persistent novelty estimators, and biologically richer signed modulation may further strengthen the practical and scientific value of this approach.

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